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PERFORMANCE EVALUATION OF YOLO MODELS FOR OBJECT DETECTION IN AUTONOMOUS VEHICLES UNDER ADVERSE WEATHER CONDITIONS

A Project/Thesis submitted in partial fulfillment for the requirement of the degree of

Bachelor of Science in

Electrical & Computer Engineering

by

Md. Robiul Islam Rabby

Roll No. 1910046

to the

Department of Electrical & Computer Engineering

Rajshahi University of Engineering & Technology

June, 2025

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Acknowledgement

This thesis has been submitted to the Department of Electrical and Computer Engineering of Rajshahi University of Engineering & Technology (RUET), Rajshahi-6204, Bangladesh, for the partial fulfillment of the requirements for the degree of B.Sc. in Electrical & Computer Engineering. The thesis is titled "**PERFORMANCE EVALUATION OF YOLO MODELS FOR OBJECT DETECTION IN AUTONOMOUS VEHICLES UNDER ADVERSE WEATHER CONDITIONS**".

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CERTIFICATE

This is to certify that the thesis entitled " ***PERFORMANCE EVALUATION OF YOLO MODELS FOR OBJECT DETECTION IN AUTONOMOUS VEHICLES UNDER ADVERSE WEATHER CONDITIONS*** " by Md. Robiul Islam Rabby (Roll No.1910046), has been carried out under my direct supervision. To the best of my knowledge, this thesis is an original one and has not been submitted anywhere for any degree or diploma.

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This is to certify that the thesis entitled " ***PERFORMANCE EVALUATION OF YOLO MODELS FOR OBJECT DETECTION IN AUTONOMOUS VEHICLES UNDER ADVERSE WEATHER CONDITIONS*** " has been corrected according to my suggestion and guidance as an external. The quality of the thesis is satisfactory.

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Abstract

This research evaluates the performance of various YOLO-based object detection models (YOLOv8x, YOLOv11n, YOLOv11s, YOLOv11m, YOLOv11l, YOLOv11x, and YOLOv12x) in the context of autonomous vehicles operating in adverse weather conditions, such as fog, rain, snow, and sandstorms. The goal is to assess the models' robustness and accuracy in environments where visibility is significantly reduced, which is critical for ensuring safe autonomous driving in real-world scenarios. The methodology includes training, validating, and testing the models on the DAWN dataset, augmented with weather-specific conditions. Performance metrics such as precision, recall, F1 score, accuracy, mAP, and mAP@50-95 were used to compare the models' performance. Larger models like YOLOv8x and YOLOv12x outperformed smaller models, showing better detection accuracy under these challenging conditions. The social impact of this work is evident in its potential to enhance the safety and reliability of autonomous vehicles. Ensuring safe operation under various weather conditions is crucial for regulatory approval and public trust in self-driving technology. Environmentally, the project contributes to the development of efficient and sustainable autonomous systems, reducing fuel consumption and emissions.

This study was motivated by the need for a balance between speed and accuracy in real-time object detection. Larger models were selected for their high accuracy, while smaller variants tested performance on resource-constrained devices. The results highlight the importance of real-world testing and continuous model optimization to improve detection accuracy in diverse environmental conditions. In the future, this research will explore multi-sensor fusion and further AI model enhancements, advancing autonomous vehicle technologies and contributing to safer, more sustainable transportation systems.

Contents

Acknowledgement iii

Certificate iv

Abstract vi

List of Figures x

List of Tables xii

List of Abbreviations xiii

1 Introduction 1

1.1 Overview 1

1.2 Background and Motivation 1

1.3 Literature Review 2

1.3.1 Case 1 2

1.3.2 Case 2 2

1.3.3 Case 3 3

1.3.4 Case 4 3

1.3.5 Case 5 4

1.3.6 Case 6 4

1.3.7 Case 7 4

1.3.8 Case 8 5

1.4 Research Gap 5

1.5 Objectives 8

1.6 Project Planning 9

1.7 Project Estimation 11

1.8 Thesis Outline 12

2 Methodology 14

2.1 Overview 14

2.2 Case Study 14

2.3 Detailed Methodology 15
2.3.1 Data Collection 16
2.3.2 Data Preprocessing 16
2.3.3 Model Selection 17
2.3.4 Training the Models 17
2.3.5 Hyperparameter Tuning 18
2.3.6 Model Evaluation 18
2.4 Justification of the Study 21
3 Design and Implementation 22
3.1 Design Overview 30
3.1.1 Hardware Design 23
3.1.2 Simulation 23
3.2 Mathematical Formulation 24
3.2.1 YOLO Architecture and Output Representation 24
3.2.2 Confidence Score Calculation 25
3.2.3 YOLO Loss Function 26
3.2.4 Backpropagation 27
3.2.5 YOLO with Adaptive Optimization Techniques 29
3.3 Implementation 30
3.3.1 Dataset Selection 31
3.3.2 Dataset Description 31
3.3.3 Preprocessing 32
3.3.4 Model Selection 34
3.3.5 Model Description 34
3.3.6 Model Implementation 39
4 Results and Discussion 41
4.1 Overview 42
4.2 Outcome of Methodology 42
4.2.1 Performance Metrics and Graphs for YOLOv11n 42

4.2.2 Performance Metrics and Graphs for YOLOv11s 46
4.2.3 Performance Metrics and Graphs for YOLOv11m 49
4.2.4 Performance Metrics and Graphs for YOLOv11l 52
4.2.5 Performance Metrics and Graphs for YOLOv11x 55
4.2.6 Performance Metrics and Graphs for YOLOv12x 58
4.2.7 Performance Metrics and Graphs for YOLOv8x 61
4.3 Comparison 64
4.3.1 Model Performance 64
4.3.2 State of Art 66
5 Social and Environmental Influence 68
5.1 Societal Impact 68
5.1.1 Financial and Health Influences 68
5.1.2 Safety, Legal and Cultural Issues 69
5.2 Impact on Environment 69
5.3 Sustainability Issues 70
6 Complex Engineering Problems and Activities 72
6.1 Addressing Complex Engineering Problems 72
6.2 Addressing Complex Engineering Activities 74
7 Conclusion and Future Plan 78
7.1 Conclusion 78
7.2 Limitations 79
7.3 Future Plan 80

Reference 82

Appendix 85

List of Figures

Figure	Figure Name	Page No.
Fig. 1.1	Thesis timeline diagram	10
Fig. 2.1	Methodology Workflow	15
Fig. 3.1	YOLOv8 basic architecture	38
Fig. 4.1	Precision graph for YOLOv11n	43
Fig. 4.2	Recall graph for YOLOv11n	43
Fig. 4.3	Precision-Recall graph for YOLOv11n	44
Fig. 4.4	F1-score graph for YOLOv11n	44
Fig. 4.5	Confusion matrix (normalized) graph for YOLOv11n	45
Fig. 4.6	Precision graph for YOLOv11s	46
Fig. 4.7	Recall graph for YOLOv11s	47
Fig. 4.8	Precision-Recall graph for YOLOv11s	47
Fig. 4.9	F1-score graph for YOLOv11s	48
Fig.4.10	Confusion matrix (normalized) graph for YOLOv11s	48
Fig.4.11	Precision graph for YOLOv11m	49
Fig.4.12	Recall graph for YOLOv11m	50
Fig.4.13	Precision-Recall graph for YOLOv11m	50
Fig.4.14	F1-score graph for YOLOv11m	51
Fig.4.15	Confusion matrix (normalized) graph for YOLOv11m	51
Fig.4.16	Precision graph for YOLOv11l	52
Fig.4.17	Recall graph for YOLOv11l	53
Fig.4.18	Precision-Recall graph for YOLOv11l	53
Fig.4.19	F1-score graph for YOLOv11l	54
Fig.4.20	Confusion matrix (normalized) graph for YOLOv11l	54

Fig.4.21 Precision graph for YOLOv11x 55
Fig.4.21 Recall graph for YOLOv11x 56
Fig.4.23 Precision-Recall graph for YOLOv11x 56
Fig.4.24 F1-score graph for YOLOv11x 57
Fig.4.25 Confusion matrix (normalized) graph for YOLOv11x 57
Fig.4.26 Precision graph for YOLOv12x 58
Fig.4.27 Recall graph for YOLOv12x 59
Fig.4.28 Precision-Recall graph for YOLOv12x 59
Fig.4.29 F1-score graph for YOLOv12x 60
Fig.4.30 Confusion matrix (normalized) graph for YOLOv12x 60
Fig.4.31 Precision graph for YOLOv8x 61
Fig.4.32 Recall graph for YOLOv8x 62
Fig.4.33 Precision-Recall graph for YOLOv8x 62
Fig.4.34 F1-score graph for YOLOv8x 63
Fig.4.35 Confusion matrix (normalized) graph for YOLOv8x 63
Fig.4.36 Result graph for YOLOv8x 64
Fig.4.37 YOLO models result comparison 65
Fig.4.38 YOLO models metric-wise result comparison 66

List of Tables

Table. 3.1	Model description	35
Table.4.1	YOLOv11n performance analysis	42
Table.4.2	YOLOv11s performance analysis	46
Table.4.3	YOLOv11l performance analysis	49
Table.4.4	YOLOv11m performance analysis	52
Table.4.5	YOLOv11x performance analysis	55
Table.4.6	YOLOv12x performance analysis	58
Table.4.7	YOLOv8x performance analysis	61
Table.4.8	All YOLO models performance analysis	65

Appendix

AV - Autonomous Vehicles

YOLO - You Only Look Once

GWO - Gray Wolf Optimizer

ARO - Artificial Rabbit Optimizer

CLEO - Chimpanzee Leader Election Optimization

CLAHE - Contrast Limited Adaptive Histogram Equalization

ESRGAN - Enhanced Super-Resolution Generative Adversarial Networks

mAP - Mean Average Precision

TP - True Positives

TN - True Negatives

FP - False Positives

FN - False Negatives

IoU - Intersection over Union

AP - Average Precision

LiDAR - Light Detection and Ranging

GPU - Graphics Processing Unit

Chapter 1

Introduction

This chapter sheds light on the rationale behind the inspiration for this thesis topic. It explores the motivation that led to choosing this specific area of research, providing an insight into the significance of the study. The following sections will cover the background, the research gaps, and the research objectives that guide this investigation.

1.1 Overview

This chapter introduces the thesis, starting with the motivation behind the research and why this topic was chosen. It then reviews the existing literature, methodologies, and key findings in the field, highlighting the gaps in knowledge that this study aims to address. The chapter also outlines the objectives of the thesis, setting the goals and direction for the research. Finally, a brief overview of the thesis structure is provided to guide the reader through the upcoming chapters and their key themes.

1.2 Background and Motivation

The growth of autonomous vehicles (AVs) promises to transform transportation globally. However, their success heavily depends on effective object detection systems that help vehicles accurately perceive their surroundings. A major challenge in real-world use is the effect of weather conditions like fog, rain, snow, and dust storms, which can weaken the performance of visual sensors such as cameras. This leads to reduced accuracy when detecting key objects like pedestrians, vehicles, and road signs. YOLO-based models, including YOLOv11 variants, YOLOv8x, and YOLOv12x, have proven to be effective for real-time object detection. This thesis explores how to improve these models' performance in harsh weather conditions. By using augmented data and enhancement techniques, the research aims to address real-world challenges and make AV perception systems more reliable and resilient, thus improving the safety and efficiency of autonomous driving technology.

1.3 Literature Review

The literature review provides an overview of the current research on object detection in autonomous vehicles (AVs) under adverse weather conditions (AWCs). As AVs are increasingly integrated into real-world environments, ensuring accurate object detection in challenging weather, such as rain, snow, fog, sandstorms, and blizzards, is vital for their safety and reliability. This section explores existing studies on object detection methods, with a focus on deep learning models like YOLO and other convolutional neural networks (CNNs). It also covers various techniques designed to enhance object detection performance in these harsh weather conditions.

1.3.1 Case 1

Object detection is a crucial task for autonomous systems, allowing them to understand their environment and make real-time decisions based on visual input. However, adverse weather conditions such as rain, snow, fog, and sandstorms severely impact the quality of images captured by sensors like cameras, LiDAR, and radar, reducing the accuracy of object detection and complicating safe navigation for autonomous vehicles (AVs).

A considerable amount of research has focused on adapting convolutional neural networks (CNNs) to enhance object detection in these challenging environments. Studies by Appiah and Mensah (2023) **[1]** and Ashraf et al. (2024) **[2]** have demonstrated that deep learning models like YOLOv5, YOLOv7, and YOLOx can perform well in certain adverse weather conditions. However, these models still face difficulties when confronted with extreme weather scenarios, such as sandstorms and blizzards.

1.3.2 Case 2

The You Only Look Once (YOLO) model is a widely used real-time object detection model, known for its speed and efficiency. Various YOLO variants, including YOLOv5, YOLOv7, YOLOx, and YOLOv9, have been evaluated in adverse weather conditions to determine their effectiveness in challenging environments.

Ashraf et al. (2024) **[2]** investigated the use of YOLOx for detecting vehicles in snow, sandstorms, rain, and fog. Their findings revealed that YOLOx-s, the smaller version of YOLOx, outperformed other YOLO variants, achieving a mAP of 0.8983 in snow and 0.8656 in sandstorm conditions. While YOLOv5 and YOLOv7 have shown effectiveness, recent developments in YOLOx variants have proven to be more robust, particularly in environments with low visibility and fluctuating lighting conditions.

1.3.3 Case 3

Research has increasingly focused on improving images affected by adverse weather conditions. Techniques like Contrast Limited Adaptive Histogram Equalization (CLAHE), Dehazing, and ESRGAN (Enhanced Super-Resolution Generative Adversarial Network) **[3]** are used to enhance images before feeding them to object detection models.

For instance, Appiah and Mensah (2023) **[1]** used ESRGAN with YOLOv7 to improve images impacted by fog and rain, resulting in better object detection accuracy. Similarly, Ashraf et al. (2024) **[2]** applied multiscale Retinex to enhance YOLOx's performance in fog and rain. These enhancement methods are essential for preparing data and ensuring models can effectively detect objects in real-world, weather-affected conditions.

1.3.4 Case 4

Hybrid approaches that combine multiple object detection models or integrate sensor data have shown promise, particularly in adverse weather. Sensor fusion has become a critical research area to improve detection accuracy in these conditions.

For instance, Zihan Chu et al. (2024) **[4]** integrated YOLOv5 with Generative Adversarial Networks (GANs) to boost the resilience of object detection. By using GAN-generated synthetic data to simulate adverse weather, their hybrid model improved detection under fog and rain. This approach enhances the system's adaptability and offers a scalable solution for autonomous systems operating in diverse weather environments.

1.3.5 Case 5

In addition to improving image quality and augmenting datasets, optimizing hyperparameters plays a crucial role in enhancing object detection performance under adverse weather. Metaheuristic algorithms like Gray Wolf Optimizer (GWO), Artificial Rabbit Optimizer (ARO), and Chimpanzee Leader Selection Optimization (CLEO) have been used to fine-tune YOLO models for better performance in these conditions.

Özcan et al. (2024) **[5]** explored the use of these metaheuristic optimization techniques with YOLOv5, YOLOv7, and YOLOv9 to improve detection in fog and rain. Their results showed that GWO and CLEO optimization led to a 6% increase in mAP for YOLOv9, demonstrating the significant impact of hyperparameter tuning on model performance in challenging weather environments.

1.3.6 Case 6

A promising research direction involves combining camera, LiDAR, and radar data to improve object detection in adverse weather. Sensor fusion enables researchers to address the limitations of individual sensors, which can be heavily affected by weather-related issues.

For example, Ashraf et al. (2024) **[6]** demonstrated that integrating camera and LiDAR data significantly enhanced vehicle detection accuracy in snowy and foggy conditions. By using YOLOx for vehicle detection and LiDAR data to refine object boundaries, this approach provided more reliable detection in low-visibility environments. This fusion method is particularly valuable for autonomous vehicles, which depend on multiple sensors to navigate and perceive their surroundings in challenging weather.

1.3.7 Case 7

Detecting small objects, such as pedestrians or bicycles, becomes particularly difficult in adverse weather conditions due to image degradation, which can obscure or misclassify these objects.

To address this, Ashraf et al. (2024) **[7]** incorporated small-object detection modules into their YOLO-based model, significantly improving the detection of small objects in low-visibility conditions. Their YOLOx system showed enhanced performance in environments where poor visibility often leads to missed detections of critical objects like pedestrians and cyclists. This approach is crucial for ensuring the safety of pedestrians and other vehicles in weather-impacted environments.

1.3.8 Case 8

A major challenge in deploying YOLO models for real-time autonomous systems is balancing detection accuracy with processing speed, especially when adverse weather conditions degrade image quality. These conditions can increase the computational load, making it difficult to maintain both speed and accuracy. Research by Ashraf et al. (2024) and Özcan et al. (2024) **[8]** shows that real-time detection can still be achieved by optimizing models and pre-enhancing images. However, the trade-off between processing time and accuracy remains a crucial issue, particularly for autonomous vehicles operating in dynamic environments.

1.4 Research Gap

Despite advancements in object detection for autonomous vehicles (AVs) under adverse weather conditions (AWCs), key research gaps still exist. These gaps hinder the development of more robust and efficient systems for real-world deployment in diverse and challenging weather environments. This section identifies these gaps, emphasizing the need for further research to improve the performance and adaptability of AVs in extreme weather conditions.

1.4.1 Insufficient Focus on Extreme Adverse Weather Conditions

A significant gap in current research is the lack of attention to extreme weather conditions, such as sandstorms, blizzards, and sun glare. While object detection in rain, fog, and snow has been extensively studied, extreme conditions like sandstorms present unique challenges. Sandstorms drastically reduce visibility, making accurate detection

difficult for YOLO-based models. Initial studies by Appiah and Mensah (2023) **[1]** and Ashraf et al. (2024) **[2]** have started exploring sandstorms, but comprehensive research remains limited. This gap is particularly important in regions like the Middle East, where sandstorms are frequent and can hinder autonomous vehicle deployment. Further research is required to optimize YOLO models for extreme weather, and specialized datasets simulating such conditions are needed, as existing datasets like DAWN mainly focus on more common weather types. Expanding these datasets will be crucial for developing more resilient models.

1.4.2 Lack of Comprehensive and Diverse Datasets

Current research often relies on datasets like DAWN, which primarily focus on fog, rain, and snow, but these datasets do not adequately cover the full range of adverse weather conditions that autonomous vehicles (AVs) may encounter. Extreme weather events, such as sandstorms and blizzards, are rarely included, limiting the ability to evaluate YOLO models under these conditions. While the DAWN dataset is valuable, it lacks the diversity required for models to generalize effectively across different adverse conditions. Moreover, few datasets combine multiple weather conditions, such as fog, rain, and snow, with accurate annotations. Ashraf et al. (2024) and Özcan et al. (2024) **[8]** emphasize that existing datasets do not fully capture the environmental challenges AVs face in real-world scenarios. To address this, new datasets are needed that incorporate a broader range of weather conditions, especially extreme events, and include real-world data from regions prone to such conditions to enable more comprehensive model evaluations.

1.4.3 Hyperparameter Optimization for AWCs

While YOLO-based models perform well in normal conditions, their effectiveness in adverse weather is limited due to suboptimal configurations. Many studies, including Özcan et al. (2024) **[5]** and Ashraf et al. (2024) **[2],** use default YOLOv5, YOLOv7, and YOLOx settings for AWCs, which are not ideal for detecting objects in challenging weather like rain, fog, snow, or sandstorms.

Metaheuristic algorithms, such as GWO, ARO, and CLEO, have been used to optimize hyperparameters for specific weather conditions. However, most research focuses on single weather types, and there is little exploration of optimizing hyperparameters across multiple conditions. Additionally, there is a lack of studies on how hyperparameters like learning rate, batch size, and layer numbers affect performance in different weather scenarios. More research is needed in hyperparameter tuning to improve YOLO models for autonomous vehicles in AWCs.

1.4.4 Need for Advanced Image Enhancement Techniques

Image enhancement techniques like CLAHE and Dehazing improve object detection in fog and rain but are less effective in extreme conditions. Super-Resolution Generative Adversarial Networks (ESRGAN), as demonstrated by Appiah and Mensah (2023) **[1],** help enhance images under poor visibility, but their effectiveness in sandstorms or blizzards has not been extensively explored. There is a clear need for more weather-specific enhancement methods to address challenges such as low contrast, motion blur, and occlusion, which are common in extreme weather. For example, sandstorms obscure image features with fine particles, while snowstorms affect lighting and texture. Future research should focus on developing enhancement techniques tailored to these conditions to improve the performance of object detection models like YOLO.

1.4.5 Real-Time Performance Under Adverse Weather Conditions

Real-time performance for autonomous vehicles (AVs) is challenging, especially with degraded images from adverse weather conditions (AWCs). While YOLO models are known for their speed, their performance can slow down when enhancement or sensor fusion techniques are applied. Ashraf et al. (2024) **[2]** and Özcan et al. (2024) **[5]** found that while these techniques improve accuracy, they also increase computational complexity. Optimizing YOLO models for both speed and accuracy under AWCs is crucial. Future research should focus on developing efficient architectures and low-latency optimizations to maintain real-time performance.

1.4.6 Multi-Sensor Fusion and Interdisciplinary Approaches

While significant progress has been made in camera-based object detection using deep learning, the combination of multiple sensor modalities remains under-explored. Each sensor type—LiDAR, radar, thermal sensors, and cameras—offers unique advantages in adverse weather. For example, LiDAR provides precise depth data, while radar excels in fog and rain, conditions where cameras struggle. Ashraf et al. and Özcan et al. (2024) **[8]** explored sensor fusion to enhance YOLO models, but multi-sensor fusion is still an emerging field in the context of AWCs. Future research should focus on multi-modal learning, enabling models to process and combine data from cameras, LiDAR, and radar for enhanced object detection. This approach could address the limitations of individual sensors and improve system reliability in adverse weather conditions.

1.4.7 Safety and Ethical Considerations in Adverse Weather

Safety and ethical concerns are crucial for autonomous vehicles (AVs) operating in adverse weather conditions (AWCs). Weather-related challenges can cause objects to be missed or misclassified, making it essential for AVs to make safe decisions in such environments. Research should explore ethical dilemmas, such as the trolley problem, where AVs must make quick decisions based on degraded sensor data. This research is vital for ensuring that AVs can navigate safely and ethically, even when their sensors are compromised by extreme weather.

In conclusion, while progress has been made in object detection for AVs in adverse weather, key gaps remain. These include the need for better datasets, hyperparameter optimization, image enhancement techniques, and real-time processing solutions. Research into multi-sensor fusion and ethical decision-making is also needed. Addressing these gaps will be vital for developing safe, reliable, and efficient autonomous systems in all weather conditions.

1.5 Objectives

The primary goal of this thesis is to improve object detection systems for autonomous vehicles (AVs) in adverse weather conditions (AWCs). The objectives are as follows:

1. **Evaluate YOLO Models:** Assess the performance of YOLO variants (YOLOv8, YOLOv11, YOLOv12) under conditions like rain, snow, fog, and sandstorms, using metrics such as precision, recall, and mAP.
2. **Investigate Image Enhancement:** Explore how image enhancement techniques (CLAHE, dehazing, ESRGAN) can improve YOLO's detection accuracy in weather-affected images.
3. **Optimize Hyperparameters:** Use metaheuristic algorithms (Gray Wolf Optimizer, Artificial Rabbit Optimizer, Chimpanzee Leader Selection Optimization) to optimize YOLO hyperparameters for better performance in real-world weather scenarios.
4. **Weather-Specific Data Augmentation:** Use synthetic data augmentation to simulate weather conditions, expanding training datasets and improving model generalization.
5. **Practical Solutions for AVs:** Recommend best practices for deploying YOLO in real-world AVs, including model selection, enhancement techniques, hyperparameter optimization, and multi-sensor integration.

1.6 Project Planning

1.6.1 Work Plan

The work plan outlines tasks for executing this research, divided into clear phases:

Phase 1: Literature Review and Data Collection

Review research on YOLO models and object detection in adverse weather. Identify key datasets (DAWN, ACDC) and organize them for preprocessing

Phase 2: Dataset Preprocessing and Augmentation

Apply data augmentation (rain, fog, snow, sandstorm effects) and image enhancement techniques (CLAHE, Dehaze). Split datasets into training, validation, and test sets.

Phase 3: Model Selection and Hyperparameter Tuning

Select YOLO variants (YOLOv8, YOLOv11, YOLOv12) for training. Evaluate initial performance with augmented datasets.

Phase 4: Model Evaluation and Refinement

Test models on the test dataset, measuring performance (precision, recall, mAP, F1 score). Refine models and hyperparameters based on results.

Phase 5: Final Evaluation and Reporting

Conduct final tests, analyze results, and write the thesis report with recommendations.

Phase 6: Submission and Presentation

Submit the thesis and prepare for the defense.

1.6.2 Timeline Diagram

The timeline diagram will visually represent the entire project's schedule, ensuring all tasks are completed on time. The following is a description of the timeline structure:

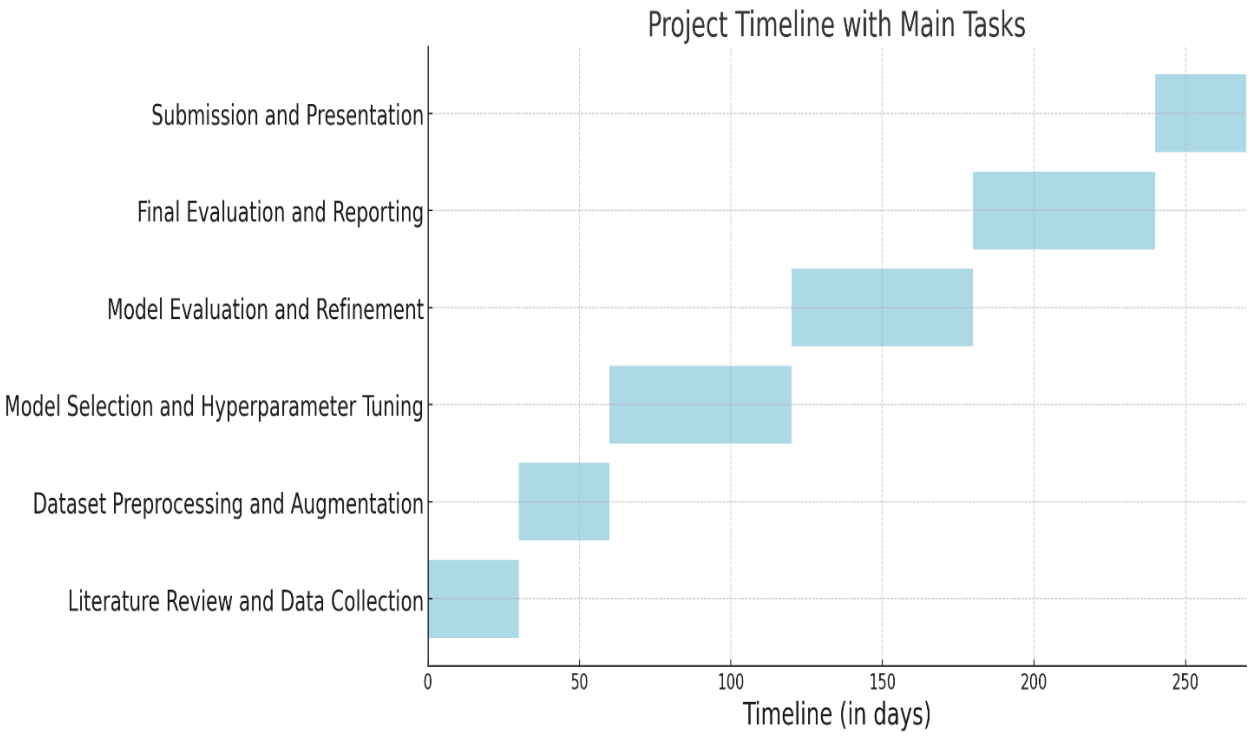


Fig. 1.1: Thesis timeline diagram

1.7 Project Estimation

1.7.1 Overall Estimation

The project is estimated to take approximately 9 months to complete, considering all stages. Below is the breakdown of the timeline:

Literature Review and Data Collection: 1 month
Dataset Preprocessing and Augmentation: 1 month
Model Selection and Hyperparameter Tuning: 2 months
Model Evaluation and Refinement: 2 months
Final Evaluation and Reporting: 2 months
Submission and Presentation: 1 month

1.7.2 Equipment Estimation

The equipment needed for this project will primarily focus on computing resources for training and testing the models. Key requirements include:

1. High-performance GPU (Graphics Processing Unit): To speed up model training, used Kaggle T4x2 GPU, especially for YOLO-based models.
2. Storage (cloud storage): To store datasets, model weights, and results.
3. Deep Learning Frameworks: PyTorch and TensorFlow for implementing YOLO models and training.

1.7.3 Financial Estimation

The financial estimation is based on the expected costs of equipment, software, and other resources. The total financial estimation for the project is as follows:

1. Kaggle Notebook: Free version ~ 0 tk.
2. Artificial intelligence support: 300*6 ~ 1800 tk.
3. Cloud storage: Google drive 100*6 ~ 600 tk

1.8 Thesis Outline

The thesis outline is divided into several portions according to get clear understanding of the thesis. They are discussed below:

Introduction

This chapter introduces object detection in autonomous vehicles (AVs), focusing on challenges in adverse weather conditions. It highlights the importance of YOLO-based models for detection, discusses the study's motivation, and provides an overview of the research objectives, methodology, and thesis structure

Literature Review

The literature review examines previous research on object detection for AVs, focusing on YOLO models and challenges posed by adverse weather conditions like rain, snow, fog, and sandstorms. It also discusses techniques such as image enhancement, sensor fusion, and data augmentation to improve detection. Gaps in existing research are identified, forming the foundation for this study's contributions.

Dataset Selection

This chapter covers the datasets used to train and test YOLO models, with a focus on the DAWN dataset. It contains images affected by weather conditions like rain, snow, and fog, and is well-suited for evaluating YOLO models' performance in real-world scenarios.

Preprocessing

This chapter outlines the preprocessing steps for the datasets, including data augmentation (motion blur, brightness adjustments, weather-specific transformations) and image enhancement techniques like CLAHE to improve image quality.

Model Selection and Implementation

This chapter describes the YOLO models selected for testing (e.g., YOLOv8, YOLOv11, YOLO12), their implementation details, and specific configurations. It also covers the fine-tuning process, including optimization with metaheuristic algorithms and custom layers to improve performance under adverse weather conditions.

Model Evaluation

This chapter presents evaluation metrics (precision, recall, mAP, F1 score) used to assess YOLO models. It discusses test results under various weather conditions and compares model performance in detecting objects in rain, snow, fog, and sandstorms.

Results and Discussion

This chapter discusses the experimental results, analyzing YOLO models' performance in adverse weather conditions. It explores the impact of image enhancement and augmentation on performance and compares the results with existing methods to highlight the improvements made in this research.

Conclusion and Future Work

The final chapter summarizes the research findings, highlighting contributions to improving object detection in AVs under adverse weather. It discusses study limitations and suggests future research directions, including developing more robust models.

Chapter 2

Methodology

In this chapter, the methodology used to carry out this research is presented. The chapter discusses the entire process of collecting data, preprocessing, model selection, evaluation, and the methods used for improving the performance of YOLO models under adverse weather conditions. The goal is to provide a clear understanding of how the research was structured and the steps taken to address the problem.

2.1 Overview

This section outlines the methodology, aimed at evaluating YOLO-based models for object detection in adverse weather conditions like rain, snow, fog, and sandstorms. The process includes dataset collection, preprocessing, model training (using YOLOv8, YOLOv11, YOLO12), and performance evaluation using metrics like mAP and F1-score. It also involves hyperparameter optimization and image enhancement techniques like CLAHE.

2.2 Case Study

This section highlights the application of YOLO models for object detection in autonomous vehicles (AVs) under adverse weather conditions (AWCs). It examines how these models perform in environments like fog, rain, snow, and sandstorms, with a focus on image enhancement, dataset augmentation, sensor fusion, and model optimization. A key study by Aloufi et al. (2023) **[9]** found that YOLOv5 outperformed YOLOv7 in sandy conditions, achieving 88% mAP in low-visibility environments. This demonstrated the importance of choosing the right model for specific weather conditions. Appiah and Mensah (2023) **[1]** used ESRGAN with YOLOv7 to enhance image quality in fog, rain, and snow, increasing detection accuracy to 80%. Meanwhile, Özcan et al. (2024) **[5]** optimized YOLOv5, YOLOv7, and YOLOv9 using metaheuristic algorithms, improving YOLOv7's mAP by 6.27%, highlighting the power of hyperparameter tuning for better performance in low-visibility conditions.

2.3 Detailed Methodology

The methodology involves key stages: data collection, preprocessing, model selection, training, hyperparameter optimization, and evaluation. These stages are outlined in detail below:

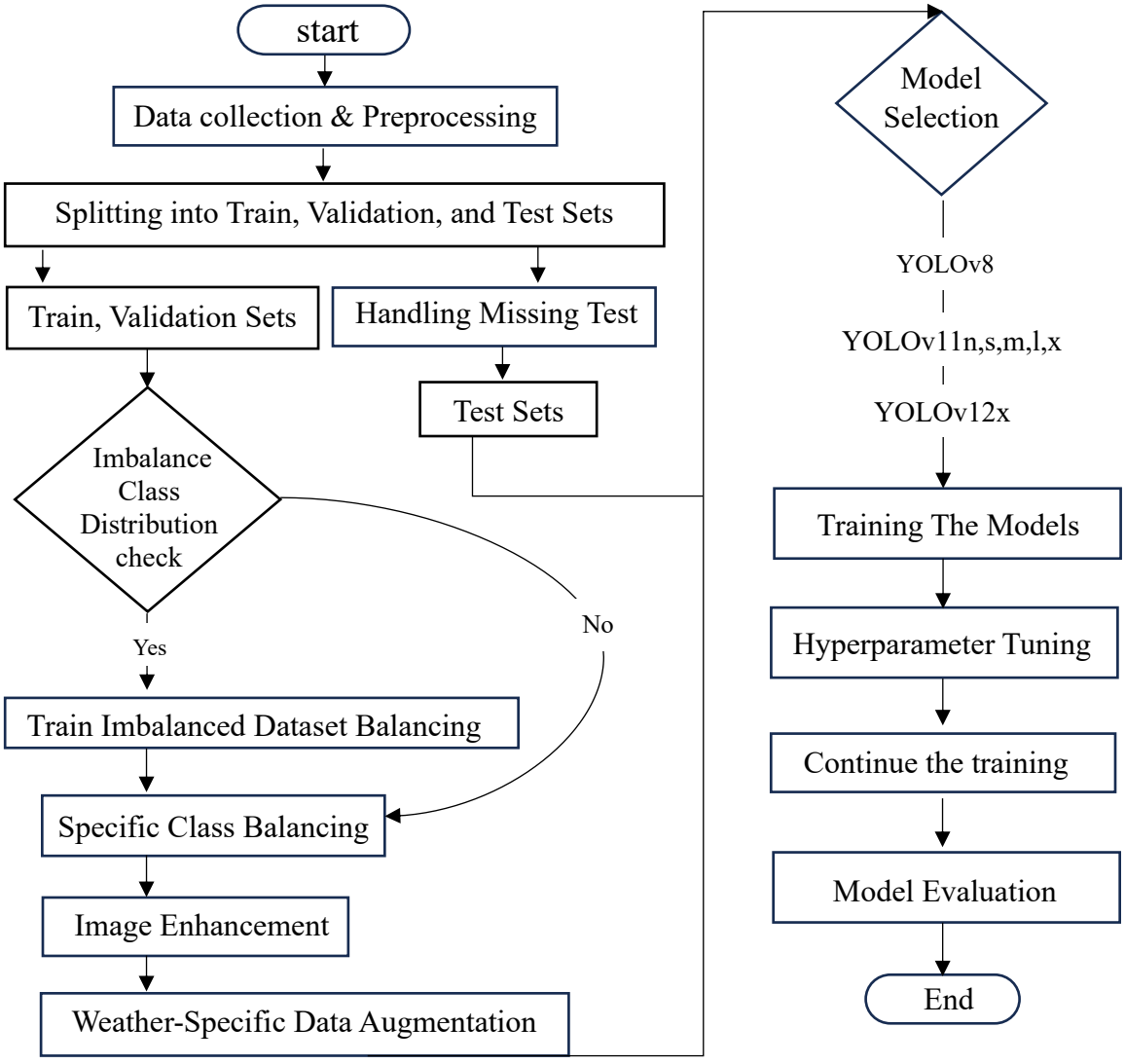


Fig. 2.1: Methodology Workflow

2.3.1 Dataset Collection

The DAWN **[10]** dataset, collected from Roboflow, is used for training and testing YOLO models. It includes 974 training images and 425 validation images, annotated for object detection in adverse weather conditions like rain, snow, sandstorms, and fog. This dataset is ideal for evaluating YOLO models in real-world, weather-impacted environments, crucial for autonomous vehicle systems.

Additional data augmentation simulates sandstorms, extreme snow, and heavy rain to assess the models' robustness in diverse conditions.

2.3.2 Data Preprocessing

Dataset selection and preprocessing are crucial for optimizing object detection models, especially for real-world challenges like detecting objects in adverse weather. This section explains how the dataset was prepared, balanced, and enhanced for training YOLO-based models.

i. **Class Imbalance and Dataset Balancing:** The first step in data preparation was inspecting the class distribution to address class imbalance, where some classes (e.g., ‘car’ or ‘person’) had more samples than others (e.g., ‘bicycle’ or ‘train’). To mitigate this, the dataset was balanced by oversampling underrepresented classes and using image augmentation to increase their instances. This ensures the model detects objects across all classes with similar accuracy, improving its generalization.

ii. **Handling Missing Data in the Test Folder:** A common issue with datasets is missing or incomplete data. During inspection, it was found that the ‘test’ folder was empty, which would hinder proper model validation. To resolve this, data from the ‘train’ and ‘valid’ folders were copied into the test set, maintaining a 70/20/10 split (70% training, 20% validation, 10% test). This standard practice ensures a balanced dataset, preventing overfitting and enabling effective training and evaluation.

iii. **Class-Specific Balancing:** The ‘train’ class had fewer data points compared to the ‘car’ class, which was overrepresented. To balance this, a 70/20/20 split was

applied to the 'train' class, adding samples through augmentation or redistribution to ensure meaningful training data.

iv. **Splitting the Dataset into Train, Validation, and Test Sets:** The data was split into training, validation, and test sets. The training set helped the model learn patterns, the validation set fine-tuned parameters, and the test set provided unbiased performance evaluation.

v. **Image Enhancement for Data Quality Improvement:** After balancing, the dataset was enhanced using techniques like CLAHE (Contrast Limited Adaptive Histogram Equalization) to improve contrast, dehazing to reduce fog and smoke, and denoising to remove noise. These methods improved image clarity, especially in low-contrast conditions.

vi. **Weather-Specific Data Augmentation:** To enhance model robustness, weather-specific augmentations like rain, fog, and snow were applied. These techniques simulated adverse weather effects, such as water droplets or reduced clarity, helping the model adapt to varying conditions. Additional augmentations like motion blur and gamma adjustments were also used to simulate dynamic weather effects.

2.3.3 Model Selection

The models selected for this research are YOLOv8, YOLOv11, and YOLO12. These variants were chosen due to their ex-effectiveness in real-time object detection tasks and their ability to handle high-resolution images. The models are trained using different configurations to test their ability to detect objects in images degraded by adverse weather conditions.

2.3.4 Training the Models

The training process fine-tunes the YOLO models on the augmented dataset using the AdamW optimizer with learning rate schedules. Models are trained for 120 epochs to ensure reliable object detection under various conditions. The loss function combines objectness, classification, and bounding box losses.

2.3.5 Hyperparameter Tuning

Hyperparameter optimization is performed using metaheuristic algorithms like Gray Wolf Optimization (GWO) **[11]** and Clever Optimization (CLEO) **[12].** These algorithms adjust key hyperparameters (learning rate, batch size, layers, activation functions) to improve model accuracy while maintaining real-time processing speed, essential for AV applications.

2.3.6 Model Evaluation

After training the YOLO models, their performance is evaluated using standard metrics to assess accuracy, minimize false positives/negatives, and ensure generalization across weather conditions. This section details the evaluation metrics, including definitions, formulas, and their role in performance assessment.

i) **Precision and Recall:** Both precision and recall are fundamental metrics in evaluating object detection models, particularly in scenarios where the balance between false positives and false negatives is crucial.

Precision measures the proportion of correct positive predictions (true positives) out of all predicted positives (both true positives and false positives):

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives+False Positives}} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

High precision means that most of the predicted objects are relevant, and the model is good at avoiding false positives.

Recall evaluates the model's ability to correctly identify all relevant objects in the dataset. It is calculated as the ratio of true positives to the total number of actual positive instances (sum of true positives and false negatives):

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives+False Negatives}} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

High recall indicates that the model successfully detects most of the actual objects, minimizing false negatives.

In object detection, precision and recall must be balanced, as improving one can decrease the other. Both metrics are reported together to assess the trade-off between false positives and false negatives.

ii) **F1-Score:** The F1-score is the harmonic mean of precision and recall, balancing their trade-off. It is particularly useful for imbalanced datasets or when both false positives and false negatives are critical.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (3)$$

The F1-score summarizes model performance by considering both false positives and false negatives. It is more informative than precision or recall alone, especially when both need to be maximized simultaneously.

iii) **Accuracy:** Accuracy is a general metric for classification models, indicating how often the model makes correct predictions. In object detection, it is the ratio of correct predictions (true positives) to total predictions (true positives, false positives, and false negatives):

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{True Positives} + \text{True Negatives} + \text{False Positives} + \text{False Negatives}}$$

$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (4)$$

- True Positives (TP): Correctly predicted objects.
- True Negatives (TN): Correctly predicted background (no object detected).
- False Positives (FP): Incorrectly predicted objects (model predicts an object that does not exist).
- False Negatives (FN): Missed objects (model fails to predict an object).

Accuracy can be misleading, especially with imbalanced datasets or when object numbers are small. For instance, a model predicting no objects (all negatives) could have high accuracy if background pixels dominate, despite poor performance.

iv) **Mean Average Precision (mAP):** Mean Average Precision (mAP) evaluates model accuracy by averaging precision across different intersection-over-union (IoU) thresholds. IoU measures the overlap between predicted and ground truth bounding boxes, with a higher IoU indicating better accuracy.

v) **Precision at different IoU thresholds:** For each object class, precision is calculated at different IoU thresholds, typically from 0.5 to 0.95 (often averaged across multiple thresholds, such as 0.5, 0.55, ..., 0.95).

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives+False Positives}} \quad \ldots\ldots\ldots\ldots(1)$$

1. **Average Precision (AP):** For each class, the average precision (AP) is calculated by averaging the precision across all IoU thresholds. The formula for AP is:

$$\text{AP} = \frac{1}{N}\sum_{i=1}^{N} \text{Precision}_i \quad \ldots\ldots\ldots\ldots\ldots\ldots (5)$$

Where N is the number of IoU thresholds.

2. **Mean Average Precision (mAP):** Finally, the mAP is the average of the APs calculated for each object class:

$$\text{mAP} = \frac{1}{C}\sum_{i=1}^{C} \text{AP}_i \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (6)$$

Where C is the number of object classes. mAP gives a single score representing the overall performance of the model across all classes and is often used for model comparison in object detection tasks.

Evaluation Under Adverse Weather Conditions: The models are evaluated under adverse weather conditions (rain, fog, snow, sandstorms) to simulate real-world challenges for autonomous vehicles, testing their robustness in reduced visibility and obscured objects.

• The evaluation is performed on a test set not used during training or validation, ensuring reliable performance metrics for unseen data.

• Comparing results from different models helps identify the best-performing model for AV systems under various weather conditions, with more robust models suited for real-world applications.

2.4 Justification of the Study

This study is motivated by the increasing demand for robust object detection systems in autonomous vehicles (AVs), especially in unpredictable and harsh weather conditions. As AVs are integrated into real-world environments, they must navigate through challenging weather phenomena like rain, fog, snow, and sandstorms, which can degrade sensor performance, particularly visual systems such as cameras. These systems are vital for object detection in autonomous driving, and the ability to function effectively in these environments is crucial for safety and reliable operation.

The need for accurate real-time detection systems in AVs is clear—vehicles must be able to identify and classify objects like pedestrians, vehicles, traffic signs, and road signals under all weather conditions. This capability is essential not only for safe driving but also for efficient navigation in dynamic, weather-affected environments. The YOLO model family is well-regarded for its real-time detection capabilities, making it an ideal candidate for this task. By evaluating its performance in adverse weather conditions, the research aims to enhance the robustness and precision of these models to operate optimally in environments where weather severely impacts visibility and sensor accuracy.

This study goes beyond testing YOLO models in ideal conditions, focusing on the unique challenges posed by adverse weather. As autonomous vehicle deployment increases, ensuring reliable performance in all environmental scenarios is critical. By optimizing YOLO models for weather-compromised environments, this research aims to contribute to the development of AV technologies that are more adaptable, reliable, and capable of operating safely in extreme weather conditions. Ultimately, this work provides valuable insights into optimizing object detection models, advancing the potential for safer and more reliable autonomous vehicles in the future.

Chapter 3

Design and Implementation

This chapter outlines the design and implementation of the YOLO-based object detection system for autonomous vehicles in adverse weather. It covers the system architecture, mathematical principles, and steps involved in dataset preparation, model selection, training, and evaluation.

3.1 Design Overview

This section provides an overview of the design framework for the YOLO-based object detection system, aimed at real-time detection in autonomous vehicles under adverse weather conditions. The system integrates high-performance hardware and deep learning models to detect and classify objects, even in fog, rain, snow, or sandstorms. The design includes key components contributing to the detection pipeline.

The development of an effective object detection system for autonomous vehicles (AVs) in adverse weather conditions involves several key steps. First, data acquisition and augmentation simulate diverse weather scenarios like fog, snow, rain, and sandstorms, preparing the models for various environmental challenges. YOLO model variants such as YOLOv8x, YOLOv11n, and others are selected and optimized to balance detection accuracy and computational efficiency for real-time operation.

The training pipeline includes advanced algorithms for model optimization, hyperparameter tuning, and regular validation to prevent overfitting. The simulation setup replicates real-world conditions to evaluate model performance in low-visibility scenarios. Lastly, hardware infrastructure is optimized to support the computational demands of deep learning models, ensuring AVs can process sensory data quickly and accurately, enabling timely decision-making in harsh weather conditions.

Each component works together to create a scalable, accurate, and real-time object detection system adaptable to changing environments. This section focuses on the hardware design and simulation aspects.

3.1.1 Hardware Design

The hardware design is crucial for optimizing the efficiency of the YOLO-based object detection system. Given the computational demands of YOLO models, especially larger variants, the hardware was selected to enhance both training and real-time inference performance. The hardware setup used for this research includes:

1. **Kaggle Free Notebooks:** The system was trained using Kaggle's free notebook environment, which provides cloud-based solutions for training machine learning models with easy dataset integration. While suitable for smaller-scale experiments, Kaggle's limited computational resources (GPUs and storage) make it less ideal for large models or datasets. **[13]**
2. **NVIDIA T4 GPUs (x2):** Two NVIDIA T4 GPUs were used for training, offering good cost-performance balance for deep learning tasks. The multi-GPU setup enabled parallel processing, accelerating the training process for YOLO models. **[14]**
3. **Memory and Storage:** Kaggle provides 30 GB of RAM and SSD storage, handling moderate datasets and high-resolution images, ensuring quick access during training and inference.
4. **Cloud-based Processing:** Kaggle’s cloud-based setup supports distributed training for larger experiments, enabling scalable operations and simplifying the management of intensive tasks like hyperparameter optimization.

3.1.2 Simulation

The simulation environment replicates real-world weather scenarios (fog, rain, snow, sandstorms) to test object detection under challenging conditions. The simulations simulate reduced visibility and occlusions, such as fog blurring objects, rain causing lens distortion,

snowflakes creating motion blur, and sandstorms adding pixel noise. These augmentations help the models learn to detect objects despite environmental degradation.

3.2 Mathematical Formulation

YOLO-based detection uses a convolutional neural network (CNN) to predict bounding boxes and class probabilities in a single pass. The model learns through gradient-based optimization and backpropagation, updating weights and biases to improve object localization, classification, and confidence estimation.

3.2.1. YOLO Architecture and Output Representation

The YOLO architecture divides the input image I (size $H \times$ W) into a grid of $S \times$ S cells. Each cell predicts B bounding boxes, class probabilities, and a confidence score.

The mathematical operations are as follows:

Bounding Box Prediction:

For each cell, YOLO predicts B bounding boxes, each with:

- x, y: Center coordinates of the bounding box relative to the grid cell.
- w, h: Width and height of the bounding box relative to the whole image.
- Confidence score: $P(\text{object}) \times \text{IoU}_{\text{pred}}^{\text{gt}}$ ... ... (7)

where IoU (Intersection over Union) is the overlap between the predicted bounding box and the ground truth.

Class Probability Prediction:

YOLO predicts the class probability $P(\text{class}_c)$ or each class ccc, outputting a vector of size C for each cell.

The total output tensor from YOLO is:

$$S \times S \times (B \times 5 + C) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (8)$$

where:

- B is the number of bounding boxes predicted per cell (usually $B = 2$).
- 5 consists of the coordinates (x, y, w, h) and the confidence score.
- C is the number of object classes (e.g., car, bus, pedestrian, etc.).

3.2.2. Confidence Score Calculation

The confidence score C_i for each bounding box is a key mathematical component. This score is calculated as:

$$C_i = P(\text{object}) \times \text{IoU}_{\text{pred}}^{\text{gt}} \ldots\ldots\ldots\ldots\ldots\ldots (7)$$

Where:

- $P(\text{object})$ is the probability that the bounding box contains an object.
- $\text{IoU}_{\text{pred}}^{\text{gt}}$ is the Intersection over Union between the predicted bounding box and the ground truth bounding box.

To calculate the IoU between two bounding boxes (predicted and ground truth), we use:

$$\text{IoU} = \frac{\text{Area of Intersection}}{\text{Area of Union}} \ldots\ldots\ldots\ldots\ldots\ldots\ldots (9)$$

Where:

- Area of Intersection: The area where the predicted bounding box and the ground truth bounding box overlap.
- Area of Union: The area covered by both the predicted and ground truth bounding boxes.

The confidence score guides the model to focus more on boxes that are more likely to contain an object, and IoU ensures that the model predicts accurate bounding box locations.

3.2.3. YOLO Loss Function

The loss function for YOLO is a composite of several terms that penalize different types of prediction errors: localization error, confidence error, and classification error. The total loss function LL combines these errors as follows:

$$L = \lambda_{\text{coord}} \sum_{i} \left(\text{IoU}_{\text{pred}} - \text{IoU}_{\text{true}}\right)^2 + \lambda_{\text{obj}} \sum_{i} \left(P(\text{object}) - \hat{P}(\text{object})\right)^2 + \lambda_{\text{noobj}} \sum_{i} \left(P(\text{no object}) - \hat{P}(\text{no object})\right)^2 + \lambda_{\text{class}} \dots\dots\dots\dots (10)$$

Where:

- **Localization Loss:** Measures accuracy of predicted bounding box coordinates.

$$\text{Localization Loss} = \sum_i \left(\text{IoU}_{\text{pred}} - \text{IoU}_{\text{true}}\right)^2 \dots\dots (11)$$

- **Objectness Loss:** Measures the model's ability to predict if a box contains an object.

$$\text{Object Loss} = \sum_i P\ (\text{object}) \log\left(\hat{P}(\text{object})\right) \dots\dots (12)$$

- **No-Object Loss:** Penalizes cells predicting an object when none is present.

$$\text{No} - \text{Object Loss} = \sum_i P\ (\text{no object}) \log\left(\hat{P}(\text{no object})\right) \dots\dots\dots\dots\dots (13)$$

- **Classification Loss:** Measures classification accuracy within bounding boxes using cross-entropy.

$$\text{Class Loss} = \sum_i$$

$$\sum_{c=1}^{C} P\left(\text{class}_c\right)\log\left(\hat{P}(\text{class}_c)\right) \ldots\ldots\ldots\ldots\ldots (14)$$

3.2.4. Backpropagation

In deep learning models like YOLO, backpropagation and gradient descent are the two fundamental operations that allow the model to learn from errors and optimize the weights of the network to improve accuracy over time.

Backpropagation is an algorithm used to compute the gradient of the loss function with respect to the weights of the neural network. This process allows the model to adjust its weights in a way that minimizes the loss function, which measures the difference between the model's predictions and the ground truth (actual values). The backpropagation algorithm works by propagating the error from the output layer back to the input layer, calculating gradients for each weight along the way. The key steps of backpropagation are:

Step 1: Forward Pass

In a forward pass, the input data (e.g., an image) is passed through the network layer by layer. Each layer performs specific mathematical operations, such as convolutions, activations, and pooling, to transform the data into predictions.

For the YOLO model, this means taking the input image, passing it through the layers of the CNN (Convolutional Neural Network), and obtaining the output in the form of predicted bounding boxes, confidence scores, and class probabilities.

Step 2: Calculate Loss

Once the predictions are made, the loss function compares the model's predictions with the ground truth values (actual labeled data). In YOLO, the loss function is a composite of the following:

- Localization loss (error in bounding box prediction).
- Confidence loss (error in predicting whether an object exists in the bounding box).
- Classification loss (error in predicting the class label).

The loss function produces a scalar value, representing the overall error of the model. The goal of training is to minimize this loss.

Step 3: Backpropagation (Calculate Gradients)

Once the loss is calculated, backpropagation begins by computing the gradient of the loss function with respect to each weight in the network. Gradients tell us how much the loss will change with a small change in the weights. The key mathematical operation here is the chain rule from calculus.

The gradient of the loss function with respect to a given weight w is computed as:

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial w} \dots\dots\dots\dots\dots (15)$$

Where:

- L is the loss function.
- o is the output of the layer (or activation function).
- $\frac{\partial L}{\partial o}$ is the derivative of the loss with respect to the output.
- $\frac{\partial o}{\partial w}$ is the derivative of the output with respect to the weight.

The chain rule helps in computing how the error in the output layer affects each of the previous layers. This process is repeated for every layer in the network, going backward from the output to the input.

Step 4: Update Weights

The gradients from backpropagation are used to update the network's weights in the direction that minimizes the error, i.e., along the negative gradient. This is achieved using an optimization algorithm like gradient descent, which adjusts the weights incrementally based on the gradients.

Learning Rate (η)

The learning rate η is a hyperparameter that controls the weight adjustment in each iteration. If too large, the model may overshoot the optimal solution.

3.2.5 YOLO with Adaptive Optimization Techniques: Adam Optimizer

YOLO often uses the Adam optimizer, an advanced variant of gradient descent that adapts the learning rate based on the first (mean) and second (uncentered variance) moments of the gradients. The update rule for Adam is as follows:

$$m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t \dots\dots\dots (16)$$

$$v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2 \dots\dots\dots\dots (17)$$

$$\widehat{m}_t = \frac{m_t}{1 - \beta_1^t} \dots\dots (18), \qquad \widehat{v}_t = \frac{v_t}{1 - \beta_2^t} \dots\dots\dots (19)$$

$$w_t = w_{t-1} - \eta \cdot \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon} \dots\dots\dots\dots\dots (20)$$

Where:

- g_t is the gradient of the loss function at time step t.
- m_t is the estimate of the first moment (mean).
- v_t is the estimate of the second moment (uncentered variance).
- β_1 and β_2 are the exponential decay rates for the moving averages (typically close to 1, e.g., $\beta_1 = 0.9$, $\beta_2 = 0.999$).

- η is the learning rate.
- ϵ is a small constant added to prevent division by zero.

The Adam optimizer adjusts the learning rate for each parameter individually based on the gradients' statistics, making it efficient for training deep models like YOLO.

a) Convergence and Stochastic Nature of YOLO

Through repeated backpropagation and gradient descent, the model updates its weights to minimize the loss function until further updates yield no significant improvement. YOLO uses stochastic gradient descent (SGD) or variants like Adam, making the optimization process noisy. This stochastic nature helps the model escape local minima and potentially find a global minimum, essential for complex object detection tasks.

b) Non-Maximum Suppression (NMS)

After predictions, Non-Maximum Suppression (NMS) removes redundant bounding boxes, retaining only the most confident predictions. The steps are:

i. Sort the predicted boxes by confidence score.
ii. Select the highest confidence box, removing others with high overlap (IoU > threshold).
iii. Repeat until only non-overlapping boxes remain.

This ensures distinct and accurate bounding boxes, minimizing false positives.

3.3 Implementation

This project involves steps like dataset selection, preprocessing, model selection, and implementing YOLO models for training, validation, and testing under simulated adverse weather conditions. This section outlines the steps for dataset preparation, model selection, and the final object detection implementation.

3.3.1 Dataset Selection

The DAWN dataset **[10]** from Roboflow was selected, designed to simulate adverse weather conditions like fog, rain, snow, and sandstorms. It contains 974 training images and 425 validation images. The initially empty test set was filled using the 70/20/10 rule for a balanced distribution, ensuring diverse weather scenarios during training and validation.

3.3.2 Dataset Description

The DAWN dataset **[10]** simulates real-world challenges for autonomous vehicles in low-visibility environments. It includes 7 object classes: Bicycle, Bus, Car, Motorcycle, Person, Train, and Truck. The images are augmented with weather conditions to test model robustness.

- **Train set:** 974 images
- **Validation set**: 425 images
- **Test set:** Data added using the 70/20/10 split

    - Bicycle, Bus, Car, Motorcycle, Person, Train, Truck

The object classes represent typical obstacles that autonomous vehicles must identify and track, ensuring the models generalize well across different scenarios.

Data Augmentation: The images are augmented with various adverse weather conditions to simulate real-world challenges, enhancing model robustness in low-visibility environments.

Data Format: The dataset is formatted for YOLO models with bounding box annotations (x_center, y_center, width, height) and includes label files with object class IDs and bounding boxes.

It is carefully prepared to ensure the models are trained, validated, and tested on data resembling real-world challenges for autonomous vehicles, making it ideal for evaluating YOLO-based models in such environments.

3.3.3 Preprocessing

Preprocessing is essential to properly format and enhance the dataset for YOLO model training. The following operations were performed to optimize the dataset, improving model performance, especially under adverse weather conditions.

i. Class Imbalance Handling

The DAWN dataset had class imbalance, with classes like 'train' and 'motorcycle' underrepresented. To address this, we oversampled underrepresented classes, ensuring the model learned effectively without overfitting to dominant classes.

ii. Creation of Test Set

The original dataset lacked a test set. Using the 70/20/10 rule, we split the data into 70% training, 20% validation, and 10% testing by copying images from the train and validation sets to maintain balance across classes.

iii. Class-Specific Augmentation for Imbalanced Classes

To improve performance on underrepresented classes (e.g., 'train'), we applied a modified 70/20/20 rule, using 20% of the training data for augmentation, enhancing performance on less frequent classes.

iv. Image Enhancement

For better performance in adverse weather, the following techniques were applied:

- **Resize for Consistency:** All images were resized to 640x640 pixels for uniform input dimensions.
- **CLAHE (Contrast Limited Adaptive Histogram Equalization):** Improved local contrast in the LAB color space, aiding feature detection in low-visibility conditions. **[15]**
- **Dehazing**: Removed haze effects caused by fog to enhance clarity.

- **Denoising:** Applied Fast Non-Local Means Denoising to reduce noise while preserving edges.
- **Sharpening:** Used a kernel filter to sharpen images and enhance object boundaries.

These enhancements were key for simulating conditions where visibility is compromised, such as rain, snow, or fog.

v. Data Split and Labeling

Once enhanced and augmented, the dataset was split into training (70%), validation (20%), and test (10%) sets. The YOLO format, which includes normalized bounding box coordinates (center, width, height), was used for labeling, ensuring compatibility with the YOLO detection framework for easier model training and evaluation.

vi. Weather-Specific Augmentation

To make the model robust to environmental changes, weather-specific augmentations were applied. The Fog Simulation reduced visibility and partially occluded objects, simulating real-world foggy conditions. Similarly, the Rain Simulation added noise and distortion to mimic water droplets on the camera lens, which obstruct visibility. The Snow Simulation introduced snowflakes to create blur and noise typical of winter weather, while the Sandstorm Simulation added dust particles to significantly reduce clarity and increase blur, replicating the challenging conditions of a sandstorm.

vii. Data Augmentation using Albumentations

Weather-specific augmentations were applied using the Albumentations library **[16],** preserving bounding box annotations. RandomGamma simulated lighting changes, while CLAHE improved local contrast for better object detection in low-visibility conditions. The OneOf function randomly applied weather-specific augmentations (Fog, Rain, Snow), exposing the model to various environmental factors. MotionBlur and GaussianNoise simulated real-world challenges like blur and noise, and Affine

Transformations ensured robustness to different object viewpoints. These augmentations helped improve the model's generalization in real-world conditions.

3.3.4 Model Selection

In this study, various YOLO variants were selected to assess their performance in object detection under adverse weather conditions. These models were chosen based on their ability to balance accuracy, real-time inference speed, and robustness in challenging environments such as fog, rain, snow, and sandstorms. The YOLOv8x model was selected for its high accuracy and fast performance, making it ideal for autonomous driving applications. YOLOv11n (Nano variant) was chosen for its lightweight design, offering faster inference and lower computational demands, making it suitable for devices with limited resources. YOLOv11s (Small variant) was selected for its balanced performance, providing a combination of speed and accuracy, ideal for real-time applications with moderate resource constraints. YOLOv11m (Medium variant) was chosen for its higher accuracy in complex tasks, offering a good balance between performance and resource usage. YOLOv11l (Large variant) was selected for high-precision applications, particularly effective in detecting large or occluded objects in complex environments. YOLOv11x (Extra-large variant) was chosen for its exceptional accuracy, ideal for complex detection tasks that require high precision. Finally, YOLOv12x (Extra-large variant) was selected for its advanced performance and efficiency, excelling in high-complexity scenarios and enhancing capabilities for detecting occlusions and blurred objects, particularly in challenging weather conditions.

3.3.5 Model Description

The YOLO models in this study are designed for real-time object detection, prioritizing speed and accuracy, especially in challenging conditions like low visibility, fog, and rain. The architecture consists of three main parts:

i) **Backbone:** The backbone extracts spatial features from input images using convolutional layers to convert raw data into feature maps. Networks like

CSPDarknet are used to detect patterns across multiple scales, ensuring robust detection in various environments.

ii) **Neck:** The neck refines and combines features extracted by the backbone, improving object detection for both small and large objects. Methods like FPN (Feature Pyramid Network) and PANet (Path Aggregation Network) are used to enhance feature fusion and maintain contextual information.

iii) **Head**: The head generates the final predictions, including class labels, bounding boxes, and confidence scores. It optimizes classification and localization through a multi-task loss function, enabling fast and accurate predictions essential for real-time autonomous driving applications.

Table 3.1: Model description

Model	Backbone	Neck	Head	Key Features
YOLOv8x	CSPDarknet	PANet (Path Aggregation Network)	Multi-task loss (bounding boxes, confidence scores, class labels)	High accuracy, excellent for both small and large objects, robust under fog, rain, and occlusion
YOLOv11n	Smaller CSPDarknet	Simplified neck	Standard YOLO head	Lightweight, fast inference, ideal for real-time applications on resource-constrained devices
YOLOv11s	CSPDarknet (with increased depth)	FPN (Feature Pyramid Network)	YOLO head (optimized for classification and bounding box prediction)	Balanced speed and accuracy, good for moderate detection accuracy in real-time scenarios

Model	Backbone	Neck	Head	Key Features
YOLOv11m	Deeper CSPDarknet	PANet	Multi-task loss	High accuracy, suitable for challenging detection tasks with more complex scenes
YOLOv11l	Deeper CSPDarknet	Enhanced PANet	Refined head for high-resolution images	High precision, ideal for detecting objects in complex environments (e.g., occlusion, clutter)
YOLOv11x	Deepest CSPDarknet	Optimized PANet	Refined YOLO head for high accuracy	Maximum detection precision, suitable for high-precision applications with complex detection tasks
YOLOv12x	Advanced backbone with multi-modal sensor fusion	PANet + FPN	Self-attention mechanisms	Advanced architecture, superior performance in real-time detection tasks under challenging weather conditions

This table summarizes the core architectural features and strengths of each YOLO model variant, comparing their backbone, neck, and head structures, along with key improvements for object detection in autonomous vehicle applications.

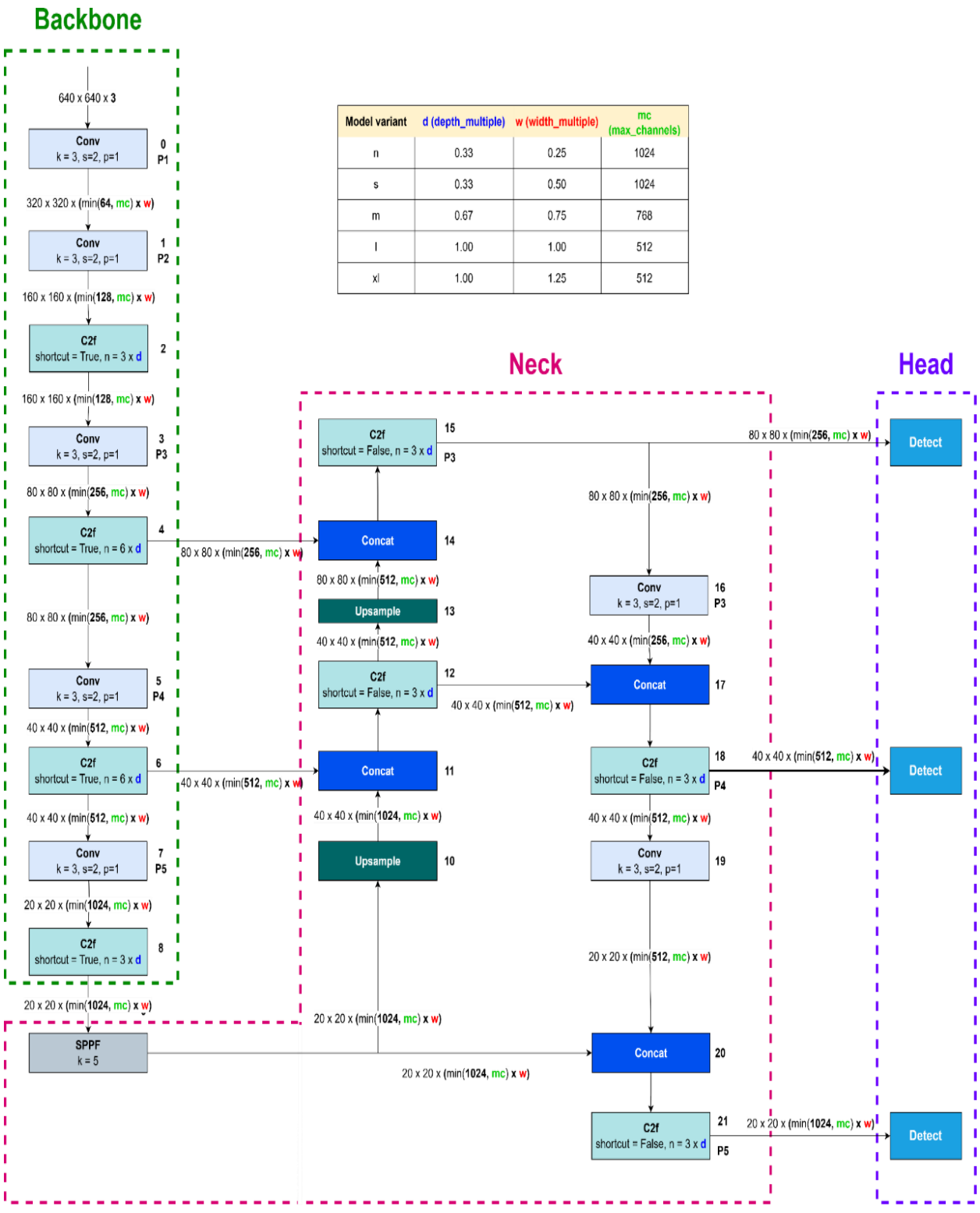
YOLOv8x Architecture:

The process of object detection in YOLO-based models involves several key stages, from input image processing to final detection outputs. First, the input image, typically resized to 640×640×3, is fed into the model for object detection. This ensures consistency across the input data, which is crucial for accurate processing and prediction. The backbone, made up of convolutional layers such as CSPDarknet, is responsible for extracting high-level features from the input image. CSPDarknet splits feature maps into branches that are then fused, enhancing both the speed and accuracy of the network. These layers detect basic elements like edges, shapes, and textures, which are critical for understanding the visual context of the image. To prevent issues like the vanishing gradient problem, residual connections are introduced, allowing gradients to flow smoothly through the network. Additionally, batch normalization stabilizes the learning process, helping the model converge faster and become more robust to different inputs.

Moving to the neck, the extracted features from different scales are aggregated. PANet, used for feature aggregation, combines low-level and high-level features through a bottom-up and top-down approach. This ensures that the model can handle objects of varying sizes. Features are then up-sampled to increase their resolution, particularly for smaller objects, and concatenated to integrate information across scales.

Finally, the head of the model makes the predictions. The bounding boxes for the detected objects are predicted, and the network classifies these objects into predefined categories. Each bounding box is assigned a confidence score, indicating the likelihood of the presence of an object. Spatial Pyramid Pooling (SPPF) applies multi-scale pooling to enhance the model's detection capabilities across various levels of detail. Non-Maximum Suppression (NMS) is then used to remove duplicate bounding boxes by keeping only the most confident predictions, based on their Intersection over Union (IoU).

The final output consists of the bounding boxes' coordinates, the class labels for the detected objects, and the confidence scores, providing the necessary information to accurately identify and locate the objects in the image.



Model variant	d (depth_multiple)	w (width_multiple)	mc (max_channels)
n	0.33	0.25	1024
s	0.33	0.50	1024
m	0.67	0.75	768
l	1.00	1.00	512
xl	1.00	1.25	512

Fig. 3.1: YOLOv8 basic architecture [17] [18]

3.3.6 Model Implementation

The model implementation was a critical part of the study, involving environment setup, model configuration, and executing the training pipeline. Below, we summarize the steps for implementing and training YOLO models for object detection under adverse weather conditions.

1. Environment Setup

The environment was set up by installing necessary libraries and frameworks, using the Ultralytics YOLO framework for model training. GPU acceleration was enabled on a Kaggle free notebook with NVIDIA T4 GPUs (x2) for efficient training.

2. Model Initialization

The YOLO variants (YOLOv8x, YOLOv11n, YOLOv11s, YOLOv11m, YOLOv11l, YOLOv11x, YOLOv12x) were initialized using pre-trained weights to leverage learned features, reducing training time and improving convergence.

3. Configuration of Hyperparameters

Key hyperparameters like learning rate, batch size, image size, and epochs were configured. An AdamW optimizer was used with a learning rate of 0.000909 and a batch size of 4. A consistent image size of 640x640 pixels and 120 epochs were chosen for training

4. Training Process

The models were trained on augmented data to simulate adverse weather conditions, helping them learn to detect objects in challenging scenarios. A multi-task loss function (classification, bounding box regression, and objectness loss) was used to enhance both classification and localization accuracy.

5. Validation and Testing

After each epoch, the models were validated on a separate validation set to track generalization. They were tested on a test set with augmented images, and performance metrics (precision, recall, F1 score, accuracy, mAP@50, mAP@50-95) were calculated.

6. Monitoring and Adjustments

Key metrics like training loss, validation loss, precision, and recall were monitored. Adjustments to the learning rate, batch size, or augmentation strategies were made if overfitting or poor performance occurred.

7. Model Saving and Results

After training, the best-performing models based on validation accuracy were saved for future use or deployment in autonomous vehicles or further experimentation.

Chapter 4

Results and Discussion

This chapter presents the performance evaluation results of various YOLO-based object detection models, tested using the augmented DAWN dataset with adverse weather conditions like fog, rain, snow, and sandstorms. The goal was to assess the models' ability to detect and classify objects in challenging environments. The models are compared based on metrics such as mAP, precision, recall, F1-score, accuracy, and inference time.

4.1 Overview

This section reviews experiments evaluating YOLO-based object detection models in adverse weather conditions to assess their performance in real-world autonomous vehicle applications, particularly in low-visibility environments like fog, rain, snow, and sandstorms. It details the experimental setup, evaluation metrics, and the strengths and weaknesses of each model.

The experiments included training, validation, and testing phases using the augmented DAWN dataset, which simulates weather conditions such as fog, rain, snow, and sandstorms. These augmentations helped enhance model robustness in realistic scenarios. The DAWN dataset, with labeled images of multiple object classes (e.g., cars, bicycles, people) under low-visibility conditions, was further augmented with the specified weather conditions to create comprehensive training, validation, and test sets.

Training Process

The training involved several steps:

- **Data Preprocessing and Augmentation:** Images were preprocessed (resized, normalized) and augmented with simulated weather effects (fog, rain, snow, sandstorms), ensuring the model could generalize to challenging conditions.

- **Model Architecture:** Various YOLO variants (YOLOv11n, YOLOv11s, YOLOv11m, YOLOv11l, YOLOv11x, YOLOv8x, YOLOv12x) were trained using an AdamW optimizer, a learning rate of 0.000909, a batch size of 4, and 640x640 input image size.
- **Epochs:** Each model was trained for 120 epochs to minimize the loss function and adjust weights.
- **Validation and Testing:** Validation was performed after each epoch to monitor the model's performance on unseen data, helping to track overfitting and ensuring the model generalized well rather than memorizing the training data.

Testing was done using a separate test dataset with adverse weather augmentations, allowing us to evaluate how well the trained models perform under real-world conditions with compromised visibility.

4.2 Outcome of Methodology

This section presents the performance evaluation outcomes of various YOLO models on the DAWN dataset, trained, validated, and tested under adverse weather conditions. The models were assessed using key metrics like precision, recall, F1-score, accuracy, and mAP.

Performance graphs were generated for each model, illustrating their behavior during training, validation, and testing. These graphs provide a visual representation of training progress and model performance, aiding in intuitive result interpretation. Detailed analysis and graphs for each model are provided in the following sections.

4.2.1 Performance Metrics and Graphs for YOLOv11n

Below is the summary of model evaluation based on precision, recall, F1-score, accuracy, and mAP metrics. The performance graph for the YOLOv11n model is included to visualize the training process and validation performance during the experiments.

Table 4.1: YOLOv11n performance analysis

Model	Precision	Recall	F1-Score	Accuracy	mAP@50	mAP@50-95
YOLOv11n	87.21%	79.93%	83.41%	67.95%	85.92%	65.27%

Graphs for YOLOv11n:

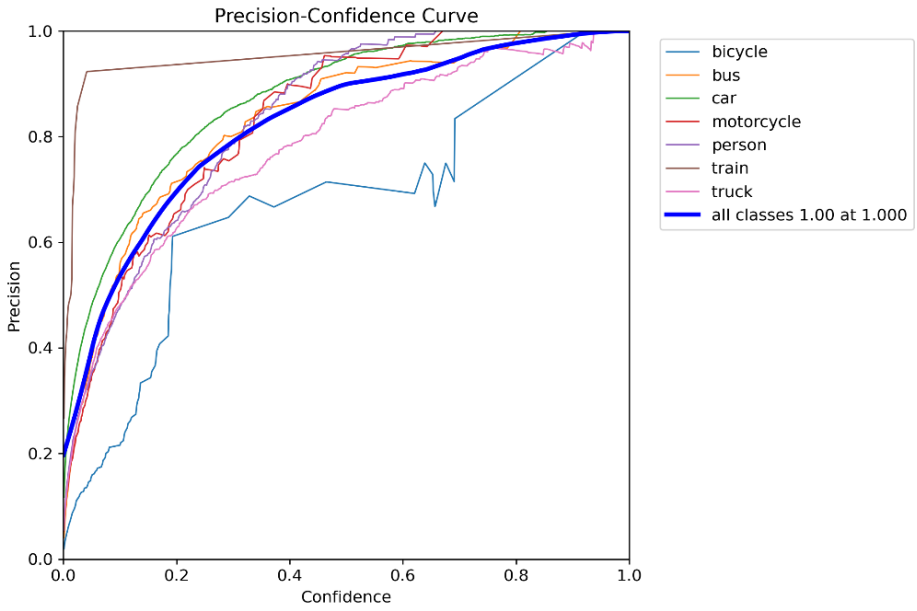


Fig. 4.1: Precision graph for YOLOv11n

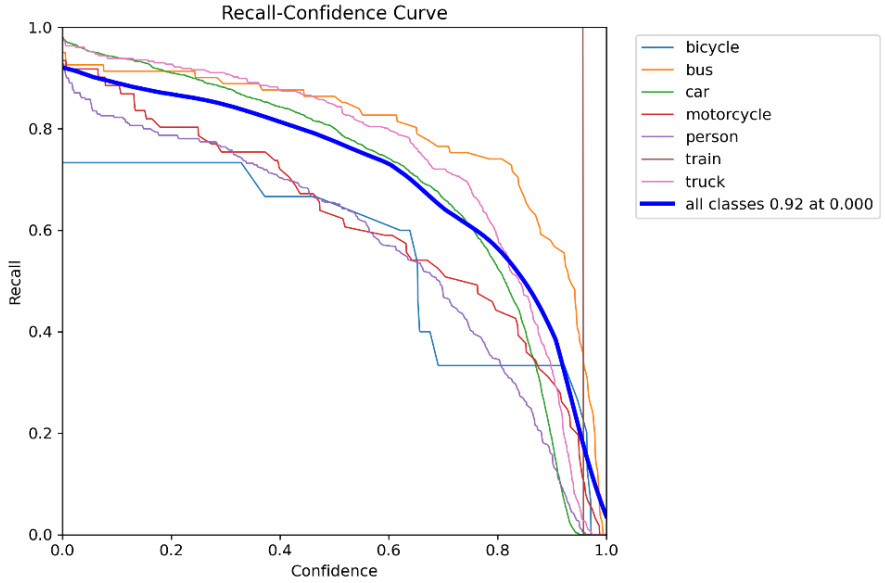


Fig. 4.2: Recall graph for YOLOv11n

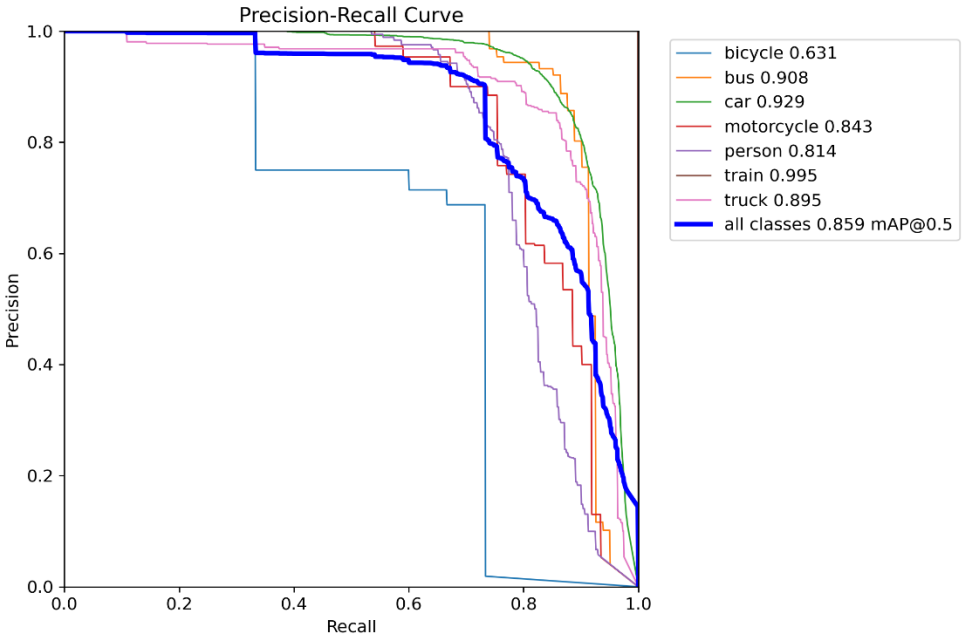


Fig. 4.3: Precision-Recall graph for YOLOv11n

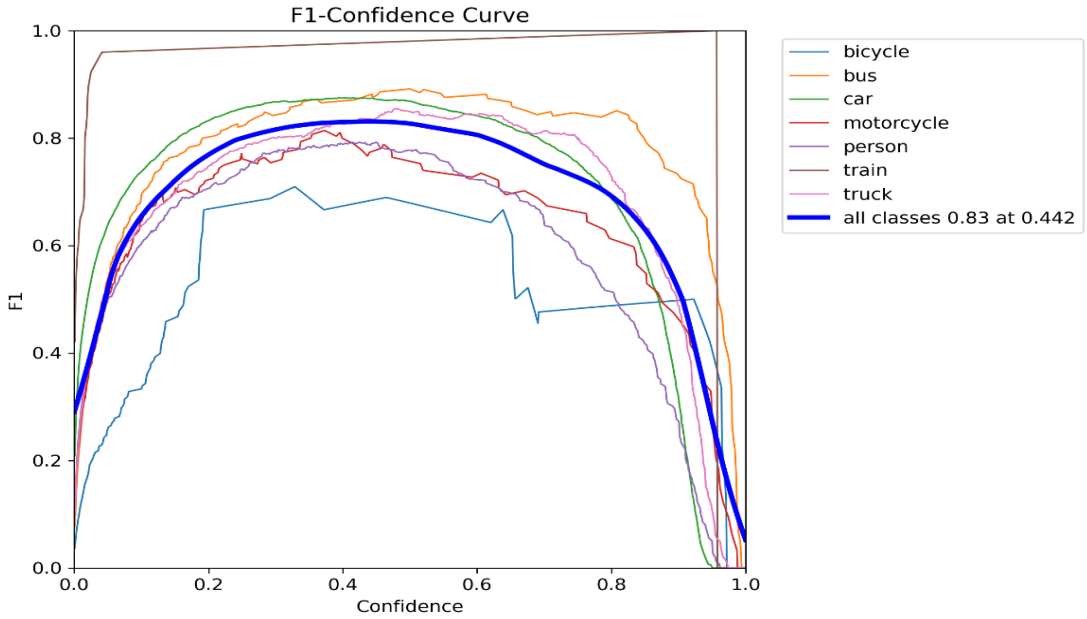


Fig. 4.4: F1-score graph for YOLOv11n

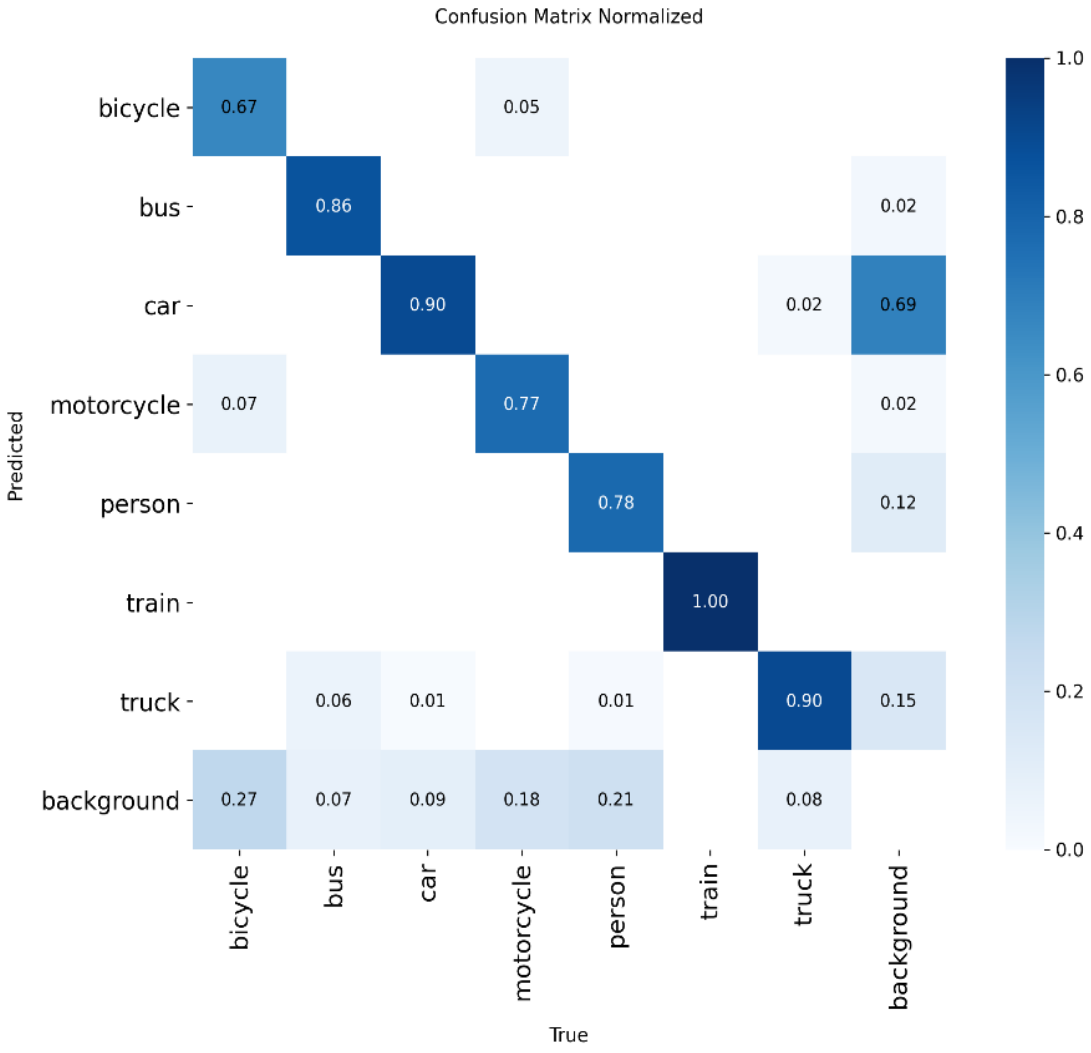


Fig. 4.5: Confusion matrix (normalized) graph for YOLOv11n

For YOLOv11n, precision remains moderate, but recall is low, indicating accurate object detection with many missed relevant objects. The training and validation loss curves suggest overfitting after some epochs, leading to poor generalization to unseen data, which explains its lower accuracy and mAP.

4.2.2 Performance Metrics and Graphs for YOLOv11s

Below is the summary of the model evaluation based on precision, recall, F1-score, accuracy, and mAP metrics. The performance graph for the YOLOv11s model is included to visualize the training and validation performance during the experiments.

Table 4.2: YOLOv11s performance analysis

Model	Precision	Recall	F1-Score	Accuracy	mAP@50	mAP@50-95
YOLOv11s	94.57%	71.61%	81.51%	68.79%	94.84%	72.94%

Graphs for YOLOv11s:

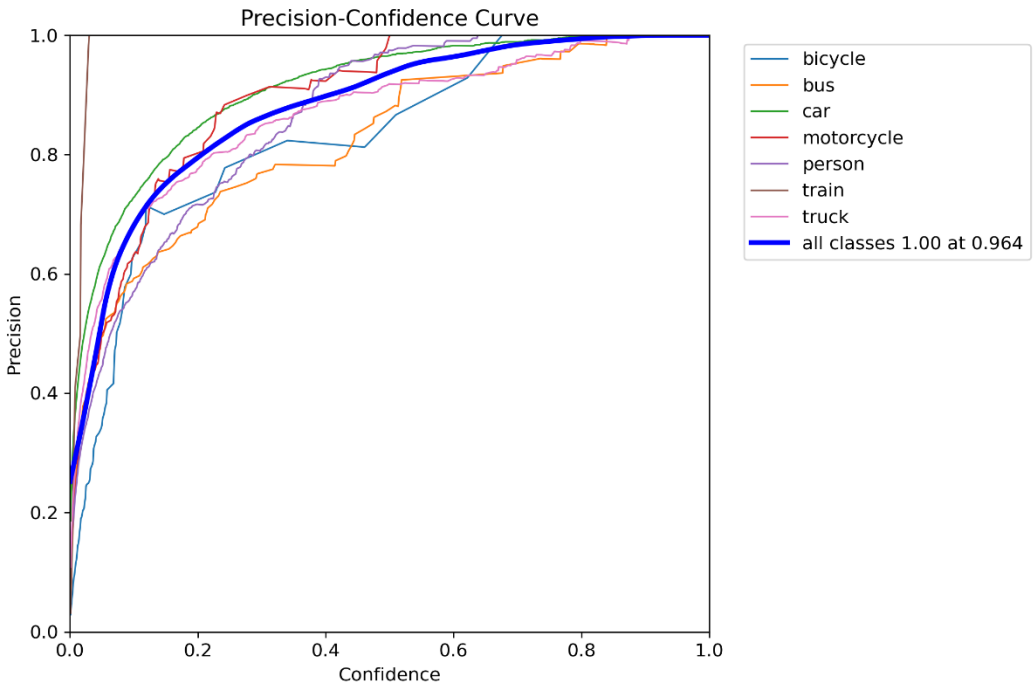


Fig. 4.6: Precision graph for YOLOv11s

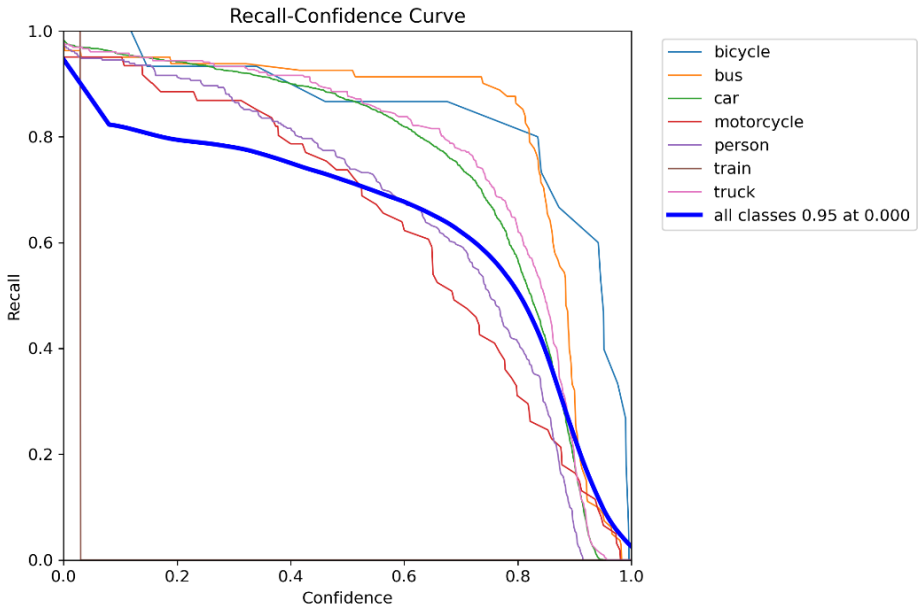


Fig. 4.7: Recall graph for YOLOv11s

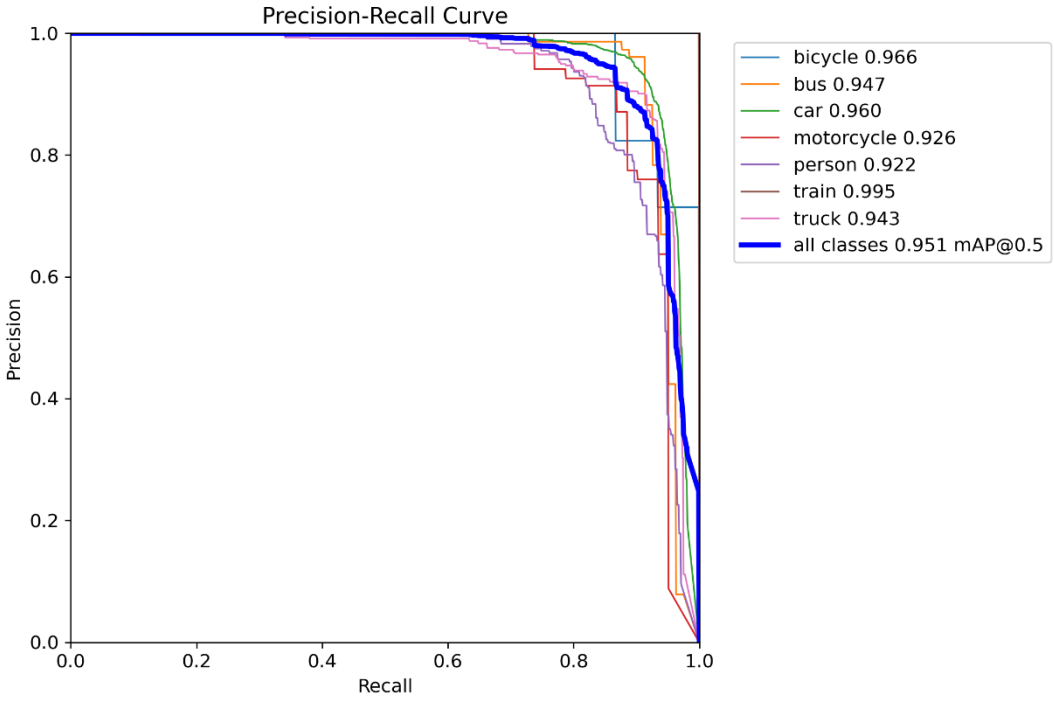


Fig. 4.8: Precision-Recall graph for YOLOv11s

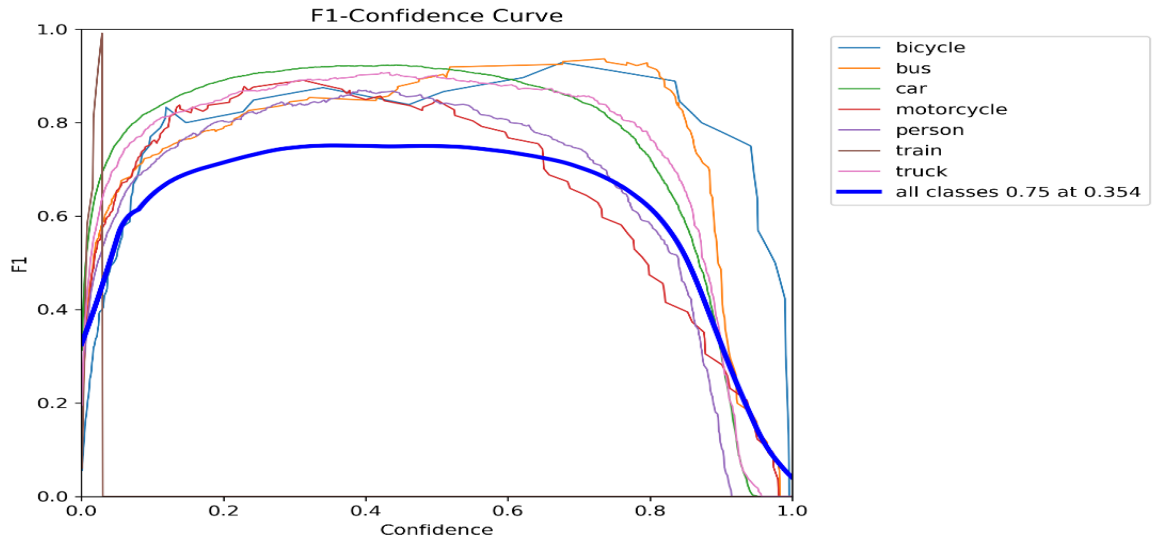


Fig. 4.9: F1-score graph for YOLOv11s

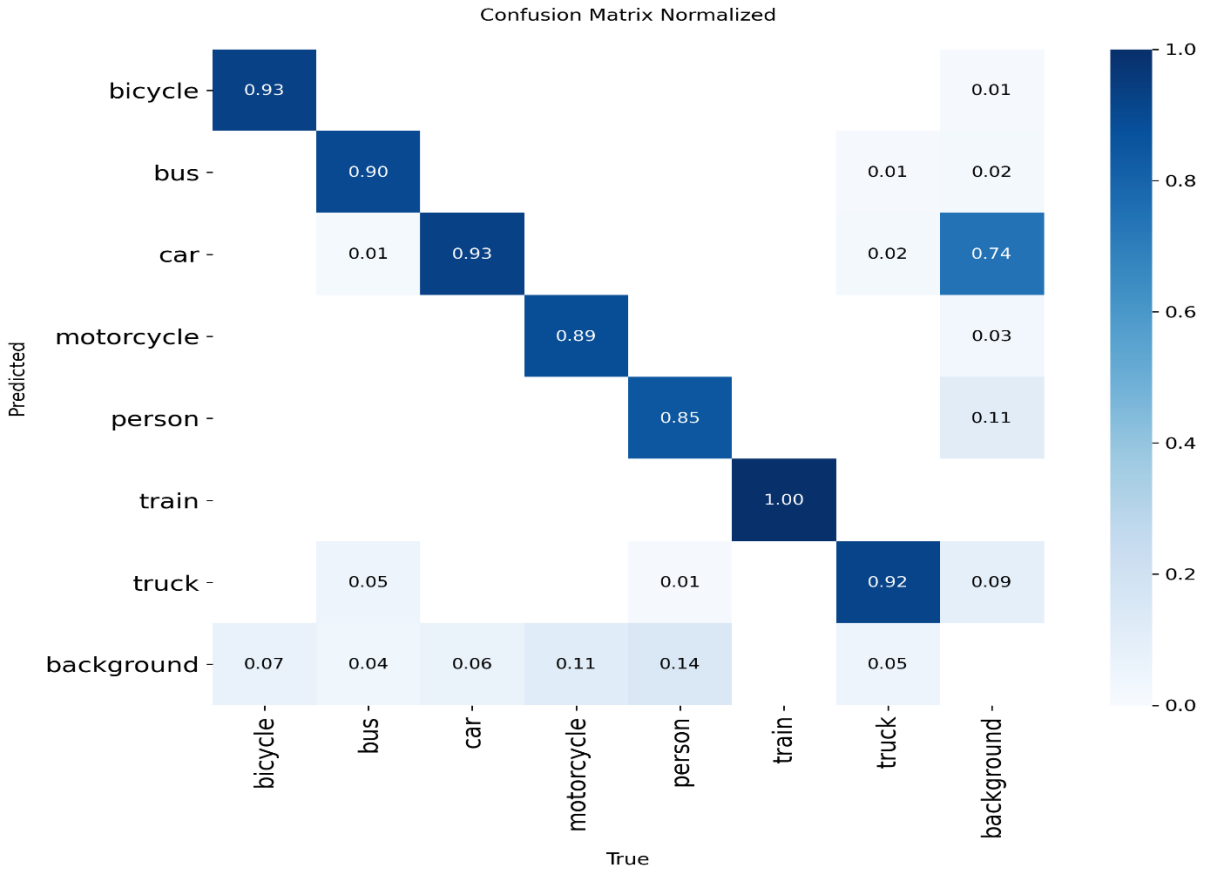


Fig. 4.10: Confusion matrix (normalized) graph for YOLOv11s

YOLOv11s demonstrates high precision, detecting objects accurately, but its low recall means it misses many objects, resulting in a good F1-score but lower accuracy and mAP.

4.2.3 Performance Metrics and Graphs for YOLOv11m

Below is the summary of model evaluation based on precision, recall, F1-score, accuracy, and mAP metrics. The performance graph for the YOLOv11m model is included to visualize the training and validation performance during the experiments.

Table 4.3: YOLOv11m performance analysis

Model	Precision	Recall	F1-Score	Accuracy	mAP@50	mAP@50-95
YOLOv11m	93.10%	76.03%	83.70%	71.97%	95.87%	74.20%

Graphs for YOLOv11m:

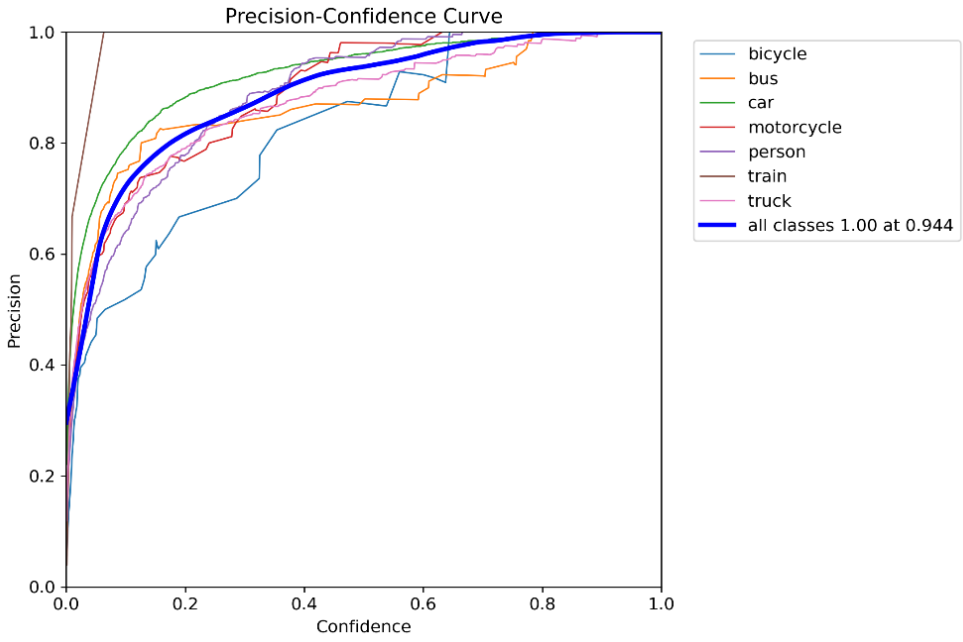


Fig. 4.11: Precision graph for YOLOv11m

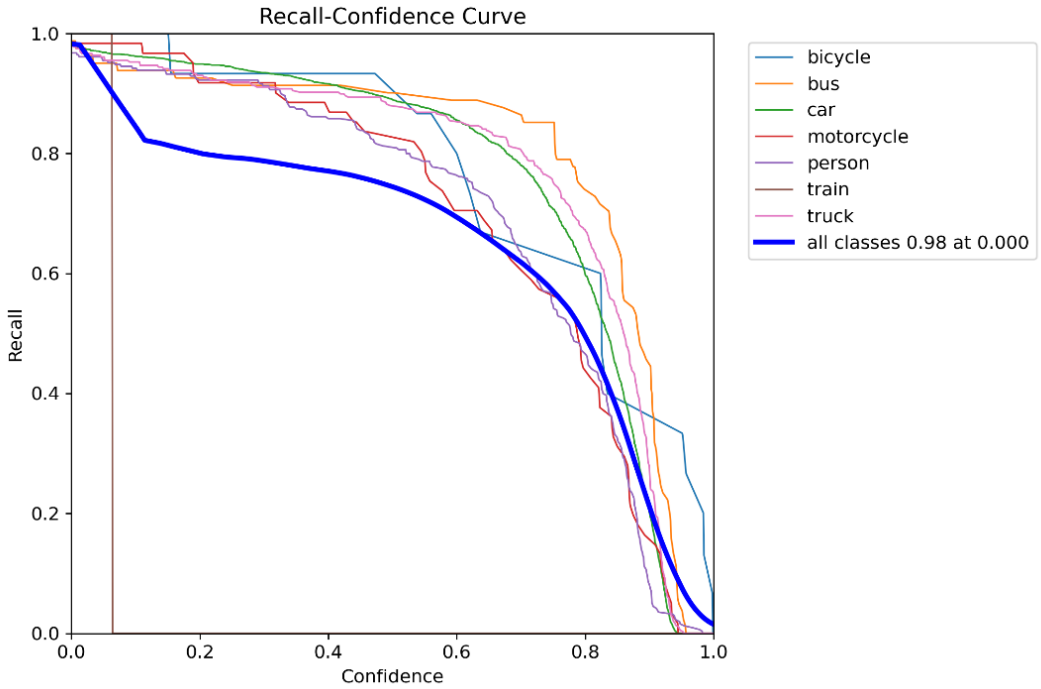


Fig. 4.12: Recall graph for YOLOv11m

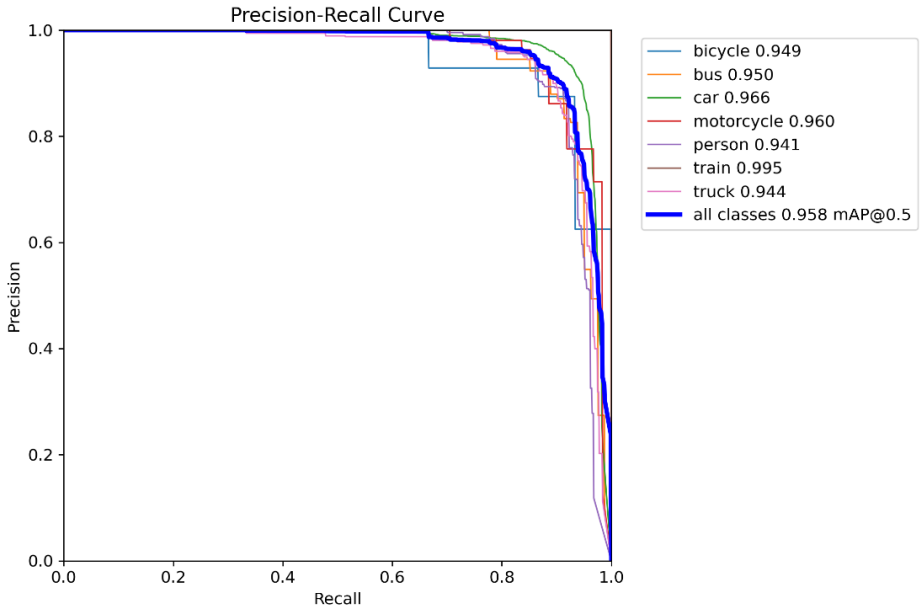


Fig. 4.13: Precision-Recall graph for YOLOv11m

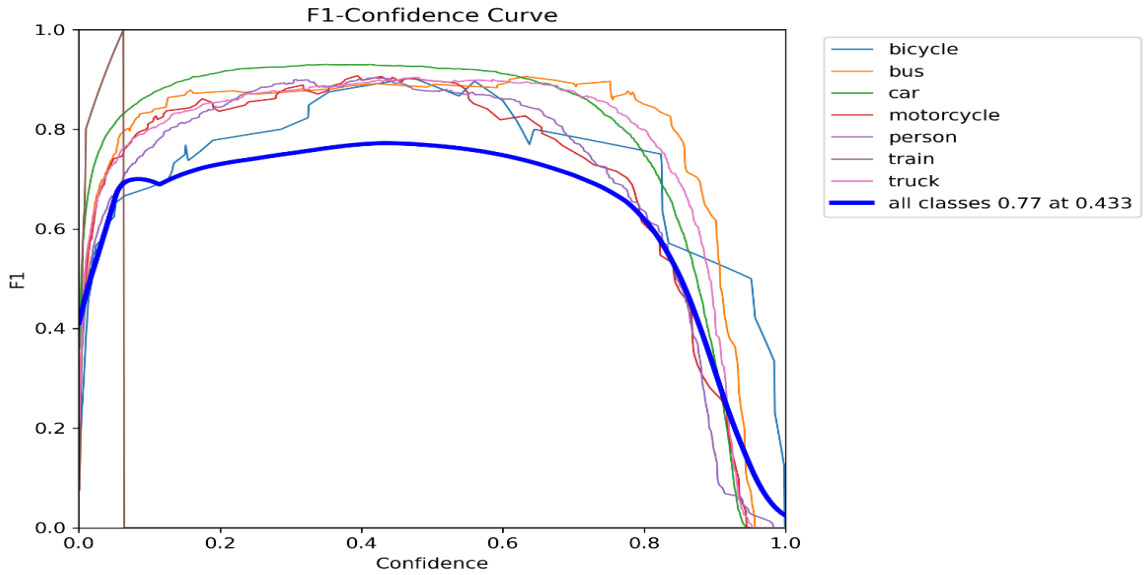


Fig. 4.14: F1-score graph for YOLOv11m

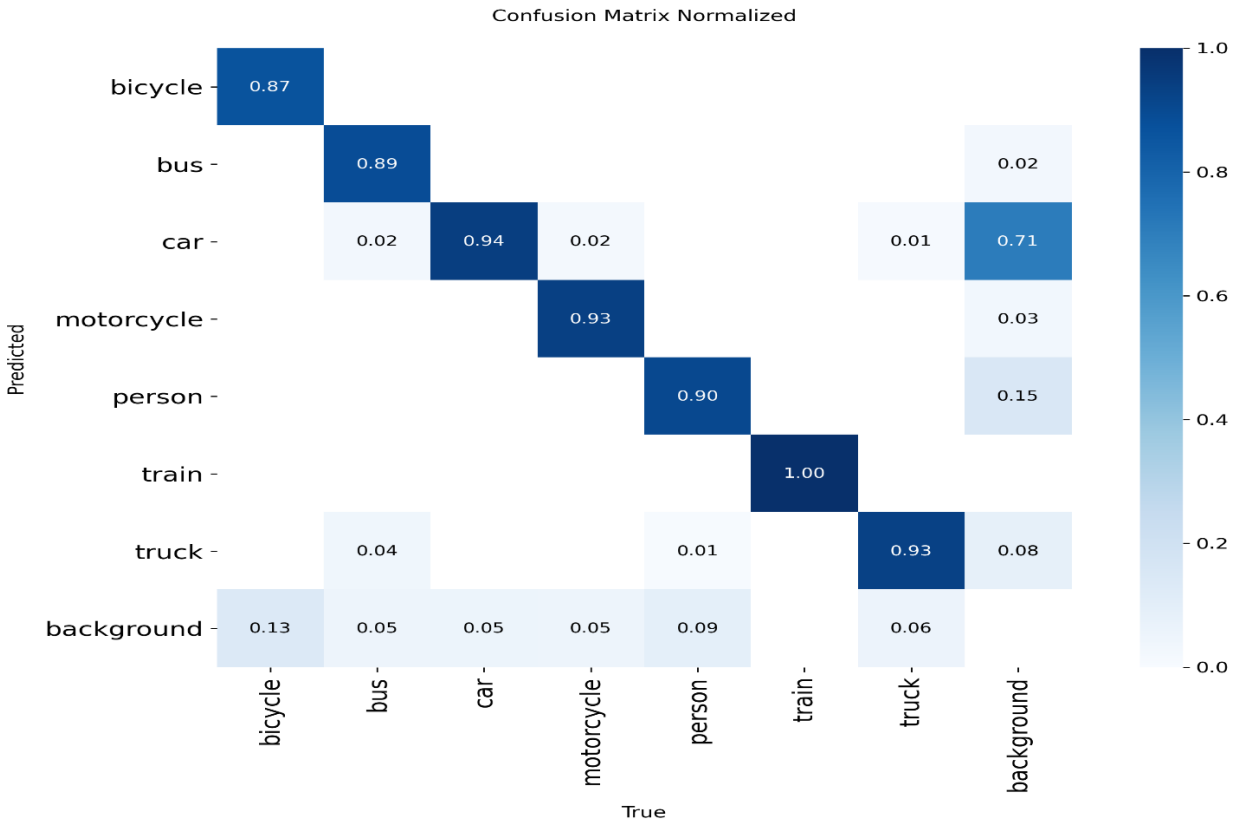


Fig. 4.15: Confusion matrix (normalized) graph for YOLOv11m

The F1-score for YOLOv11m indicates a good balance between precision and recall. While its precision and recall are slightly lower than YOLOv8x and YOLOv11x, it still performs well.

4.2.4 Performance Metrics and Graphs for YOLOv11l

Below is the summary of model evaluation based on precision, recall, F1-score, accuracy, and mAP metrics. The performance graph for the YOLOv11l model is included to visualize the training and validation performance during the experiments.

Table 4.4: YOLOv11l performance analysis

Model	Precision	Recall	F1-Score	Accuracy	mAP@50	mAP@50-95
YOLOv11l	93.17%	90.61%	91.87%	84.96%	95.40%	77.16%

Graphs for YOLOv11l:

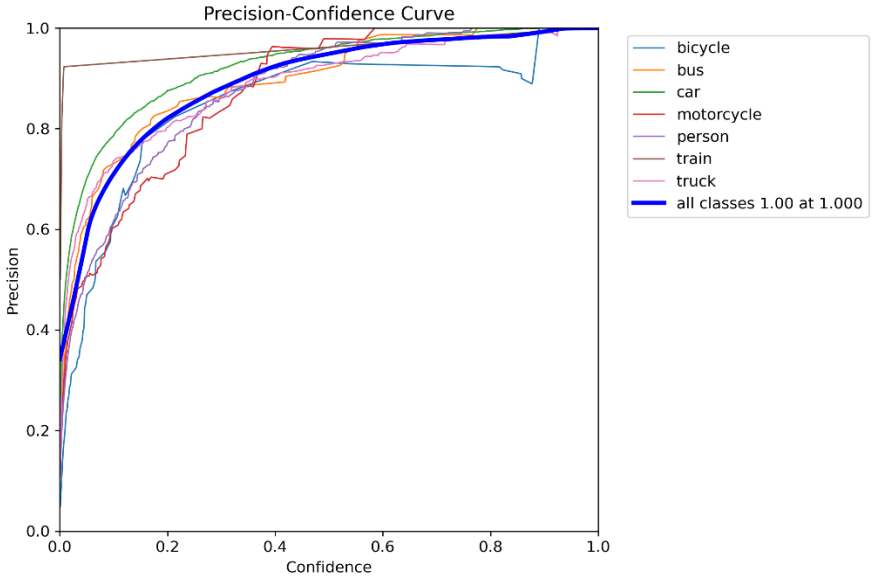


Fig. 4.16: Precision graph for YOLOv11l

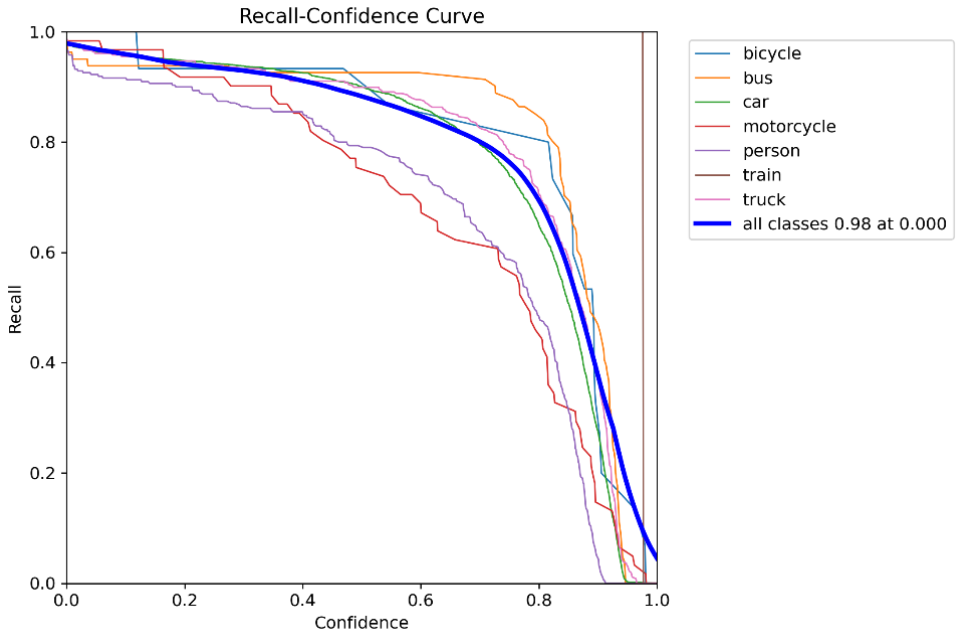


Fig. 4.17: Recall graph for YOLOv11l

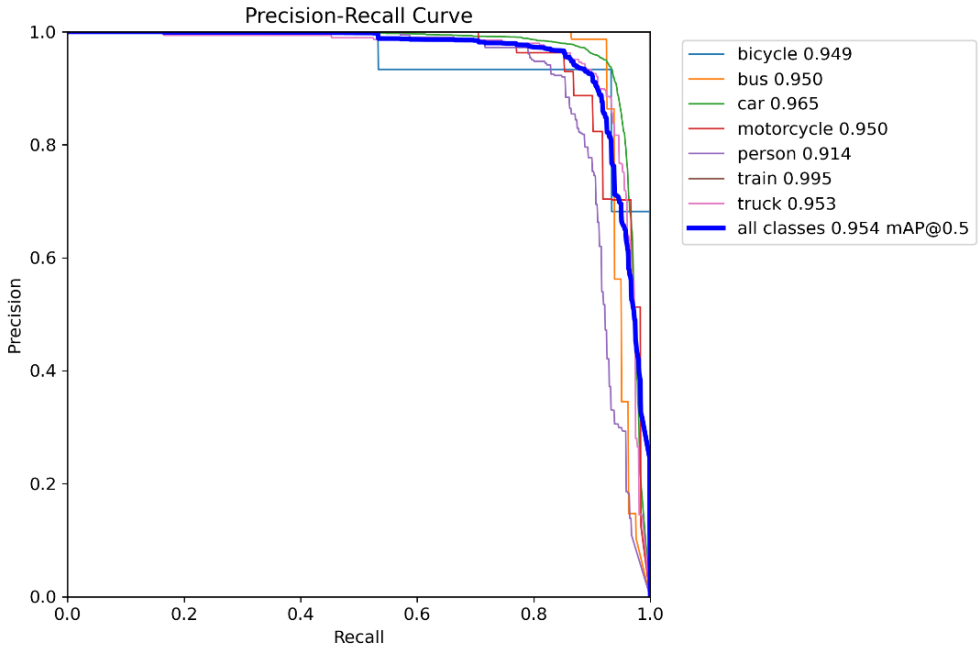


Fig.4.18: Recall graph for YOLOv11l

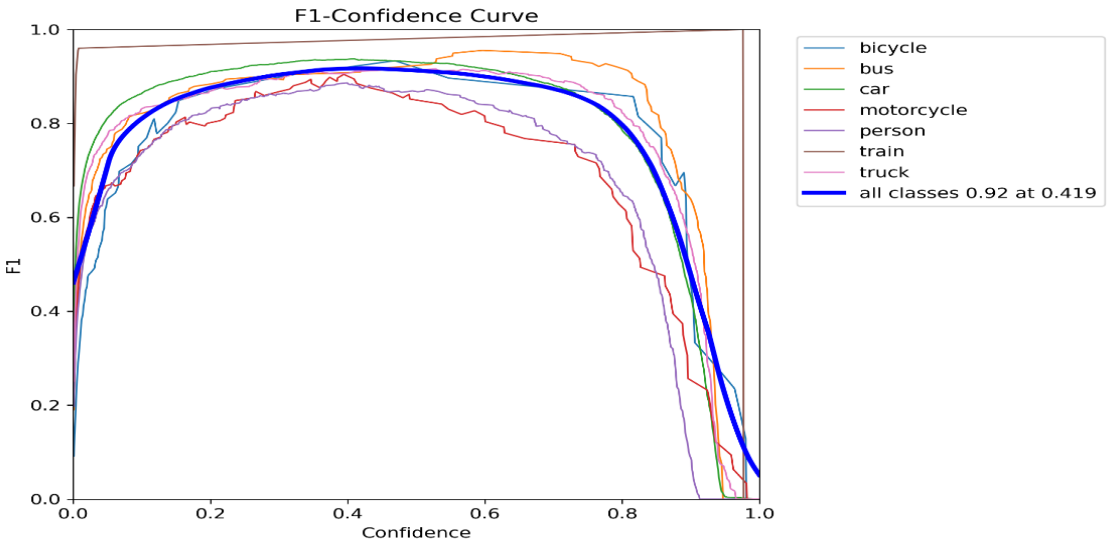


Fig. 4.19: F1-score graph for YOLOv111

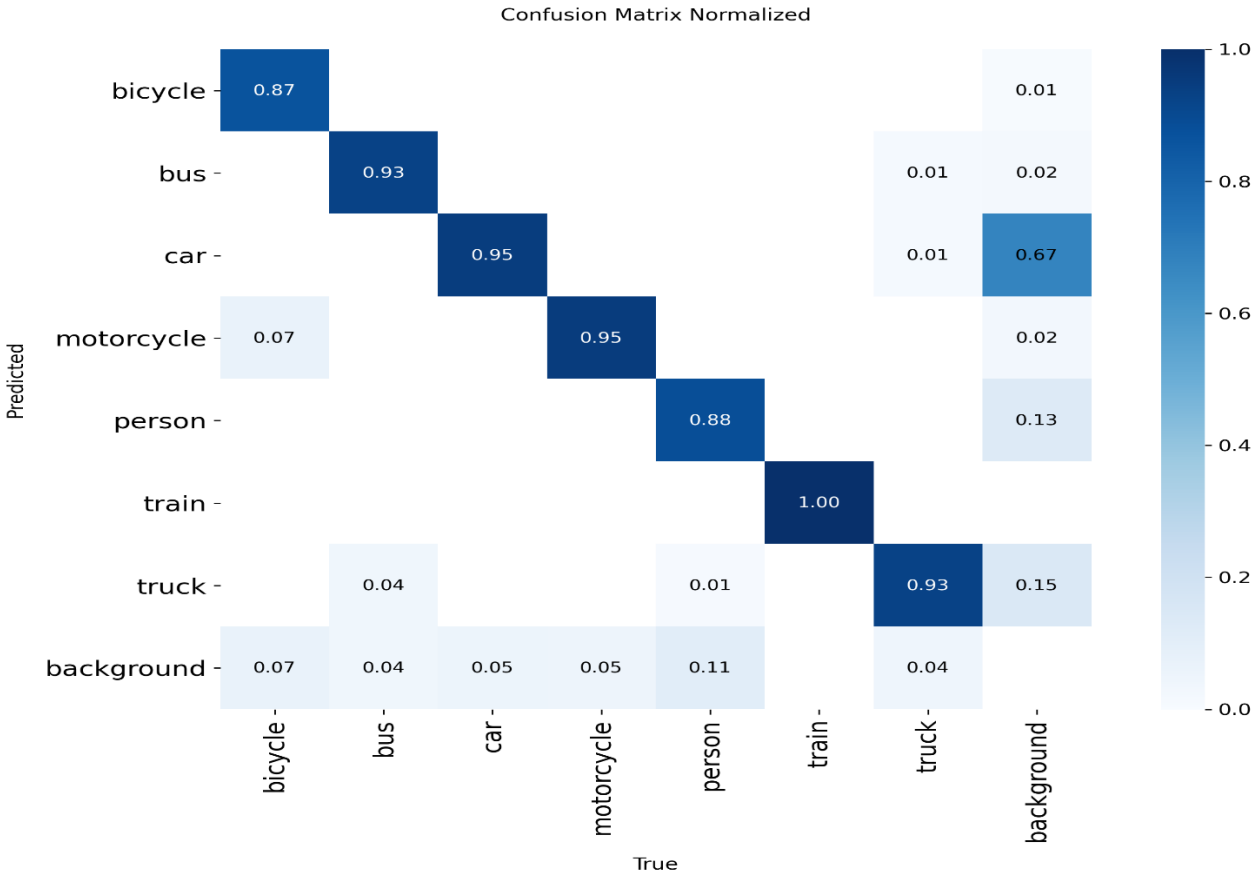


Fig. 4.20: Confusion matrix (normalized) graph for YOLOv111

YOLOv11l performs exceptionally well with high recall (90.61%) and strong precision, indicating its effectiveness in detecting most objects while maintaining accuracy.

4.2.5 Performance Metrics and Graphs for YOLOv11x

Below is the summary of model evaluation based on precision, recall, F1-score, accuracy, and mAP metrics. The performance graph for the YOLOv11x model is included to visualize the training and validation performance during the experiments.

Table 4.5: YOLOv11x performance analysis

Model	Precision	Recall	F1-Score	Accuracy	mAP@50	mAP@50-95
YOLOv11x	93.36%	90.91%	92.12%	85.39%	96.08%	76.63%

Graphs for YOLOv11x:

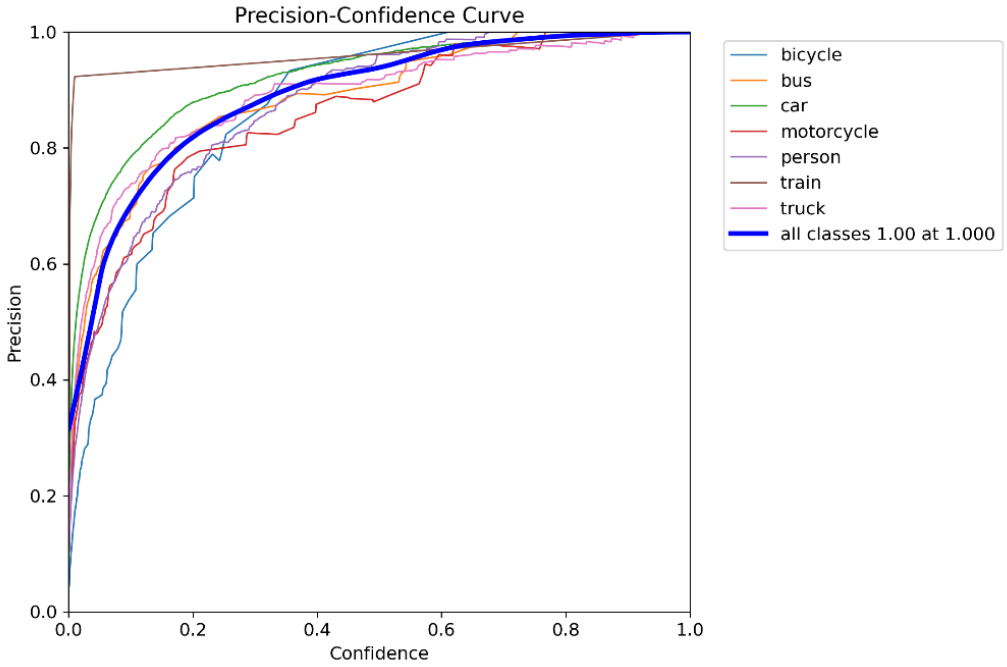


Fig. 4.21: Precision graph for YOLOv11x

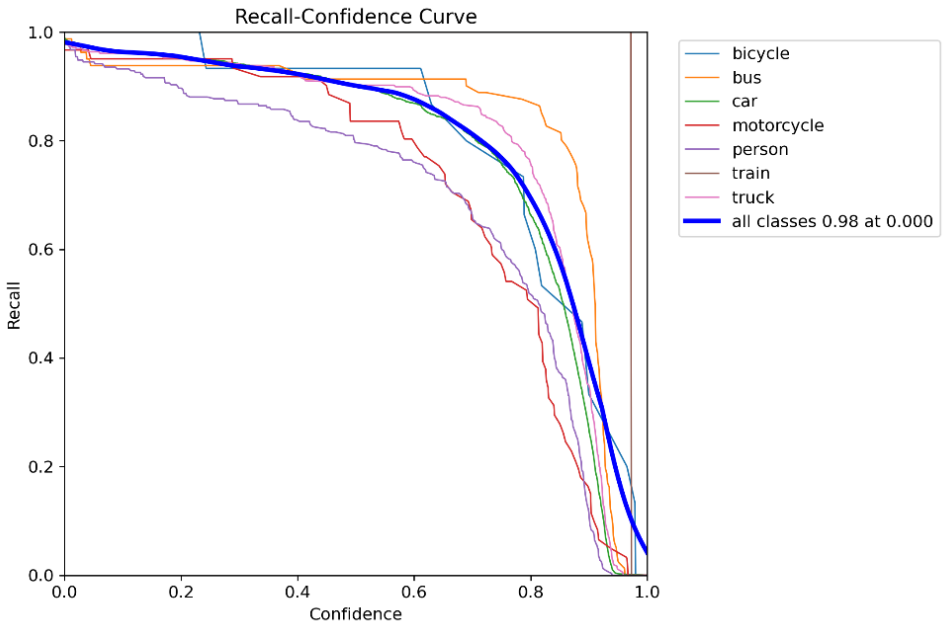


Fig. 4.22: Recall graph for YOLOv11x

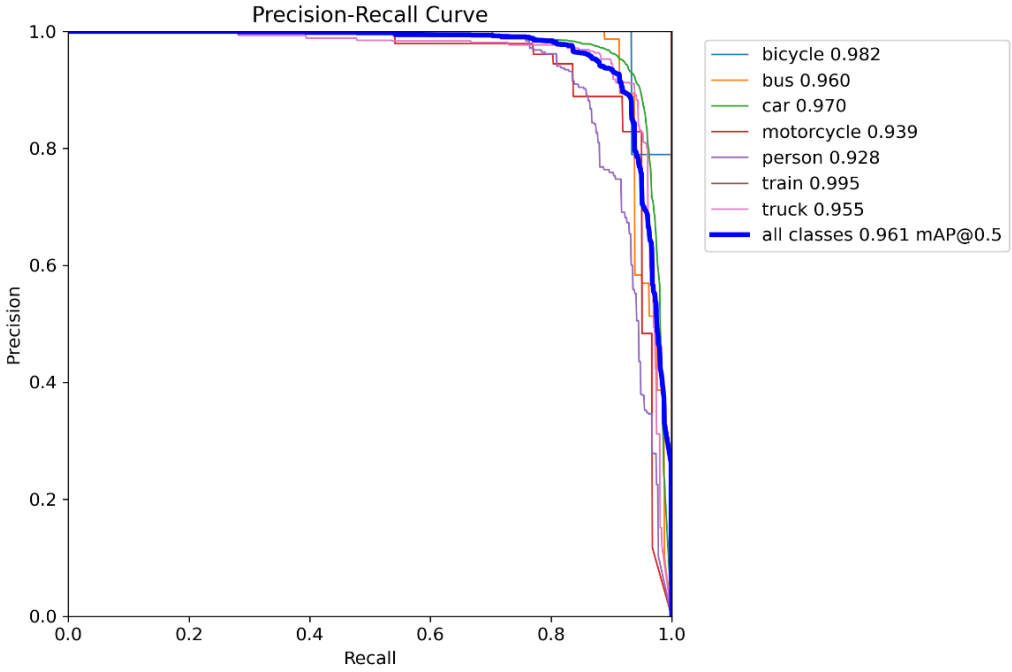


Fig. 4.23: Precision-Recall graph for YOLOv11x

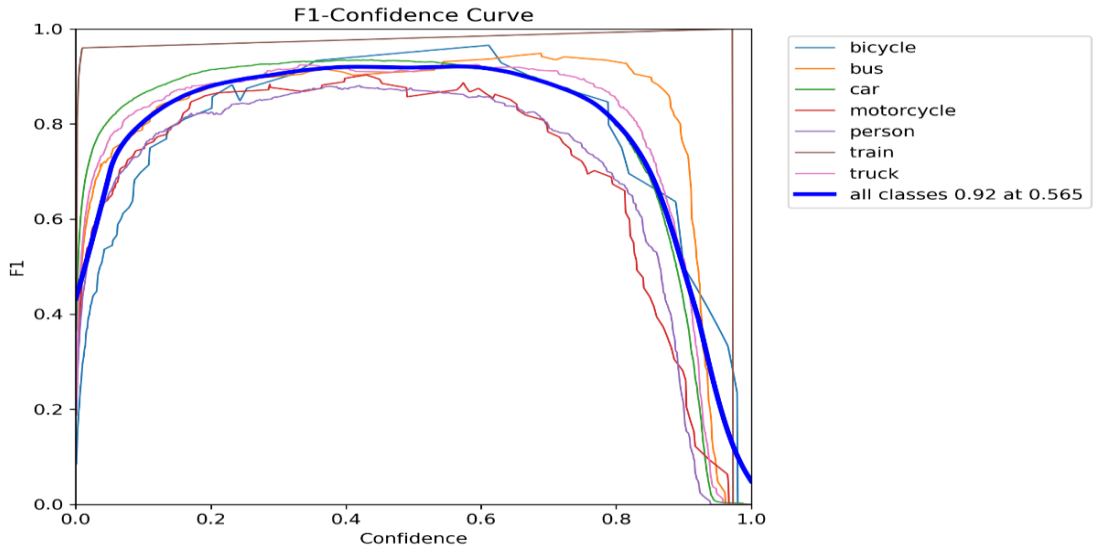


Fig. 4.24: F1-score graph for YOLOv11x

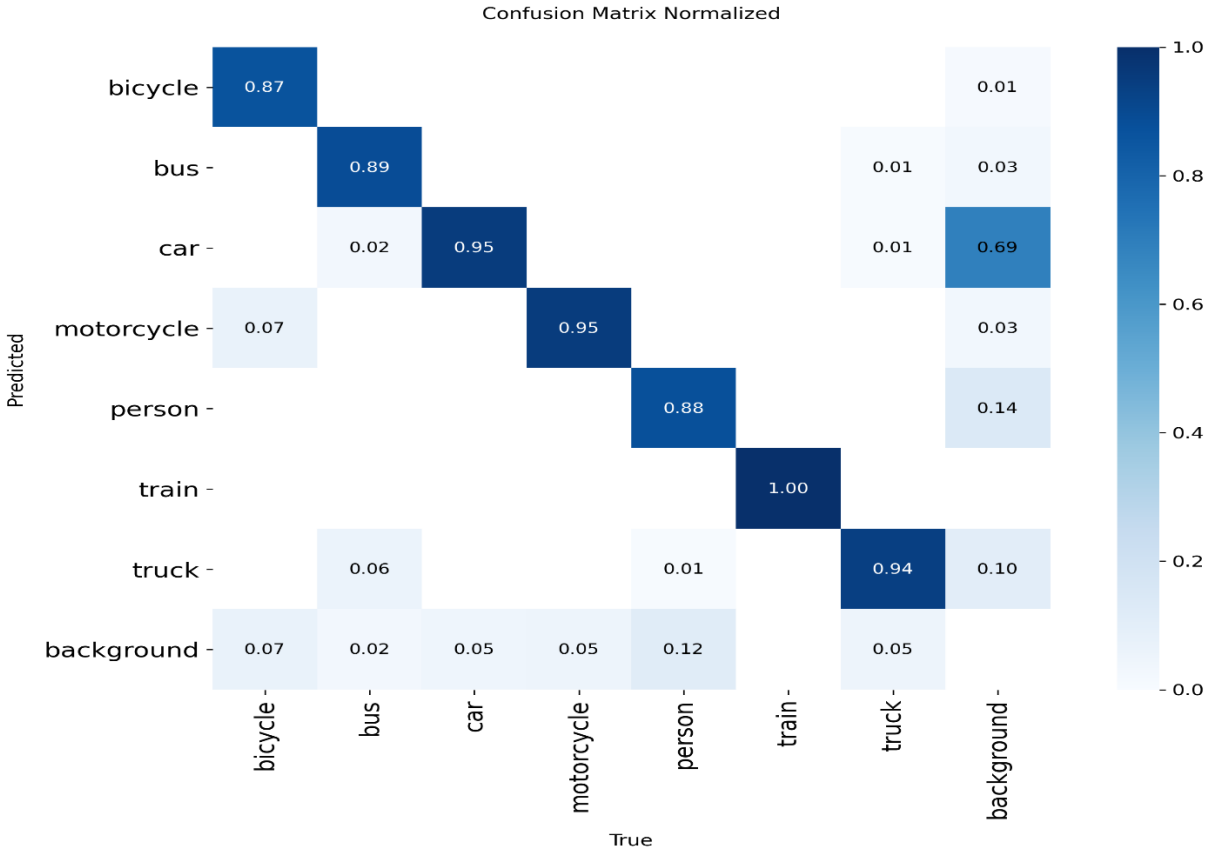


Fig. 4.25: Confusion matrix (normalized) graph for YOLOv11x

YOLOv11x exhibits high precision and recall, with a good balance between detecting objects accurately and finding most of the objects in the scene. It performs nearly as well as YOLOv8x but slightly lags in terms of overall mAP and accuracy

4.2.6 Performance Metrics and Graphs for YOLOv12x

YOLOv11x shows high precision and recall, balancing accurate object detection with identifying most objects. While it performs nearly as well as YOLOv8x, it slightly lags in mAP and accuracy.

Table 4.6: YOLOv12x performance analysis

Model	Precision	Recall	F1-Score	Accuracy	mAP@50	mAP@50-95
YOLOv12x	91.70%	87.34%	89.47%	80.94%	94.30%	74.58%

Graphs for YOLOv12x:

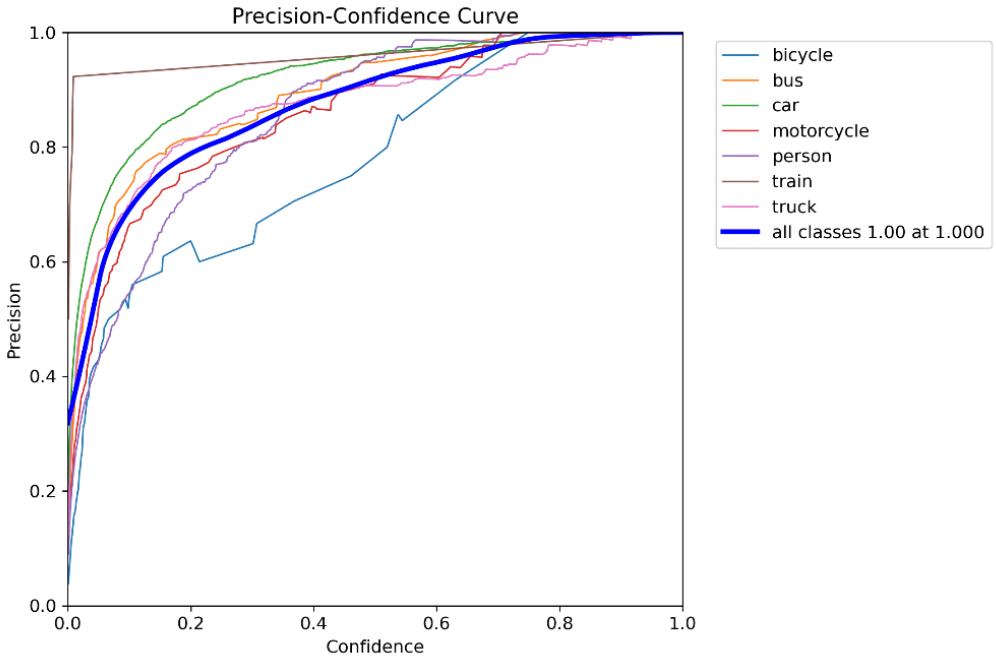


Fig. 4.26: Precision graph for YOLOv12x

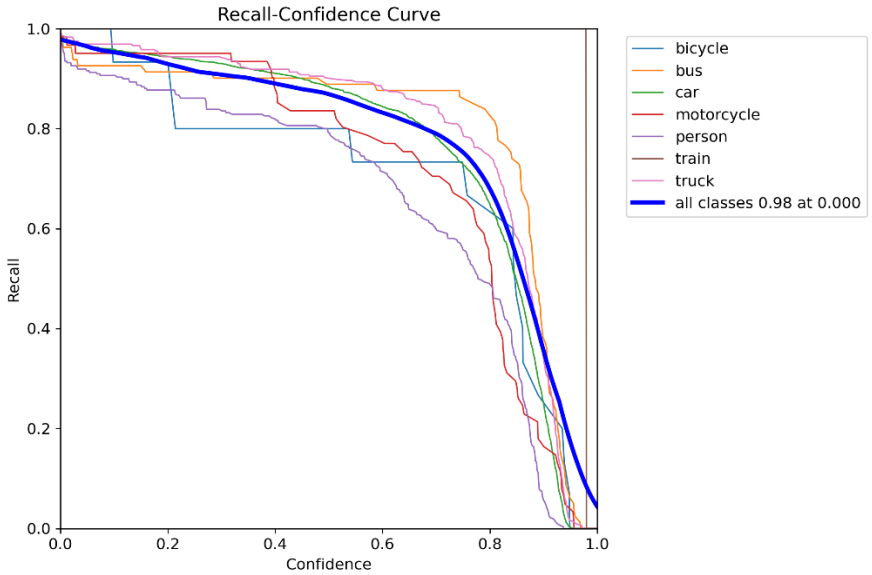


Fig. 4.27: Recall graph for YOLOv12x

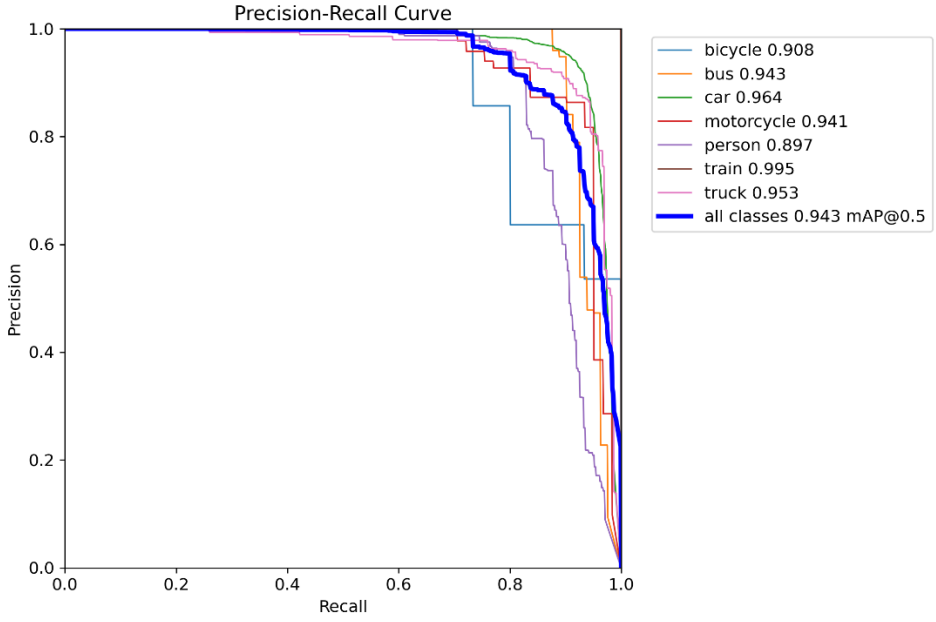


Fig.4.28: Precision-Recall graph for YOLOv12x

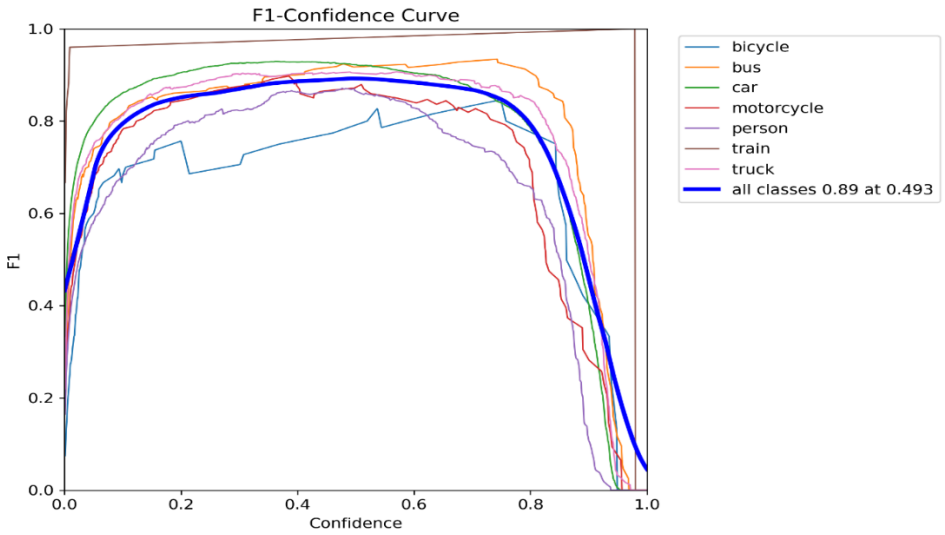


Fig. 4.29: F1-score graph for YOLOv12x

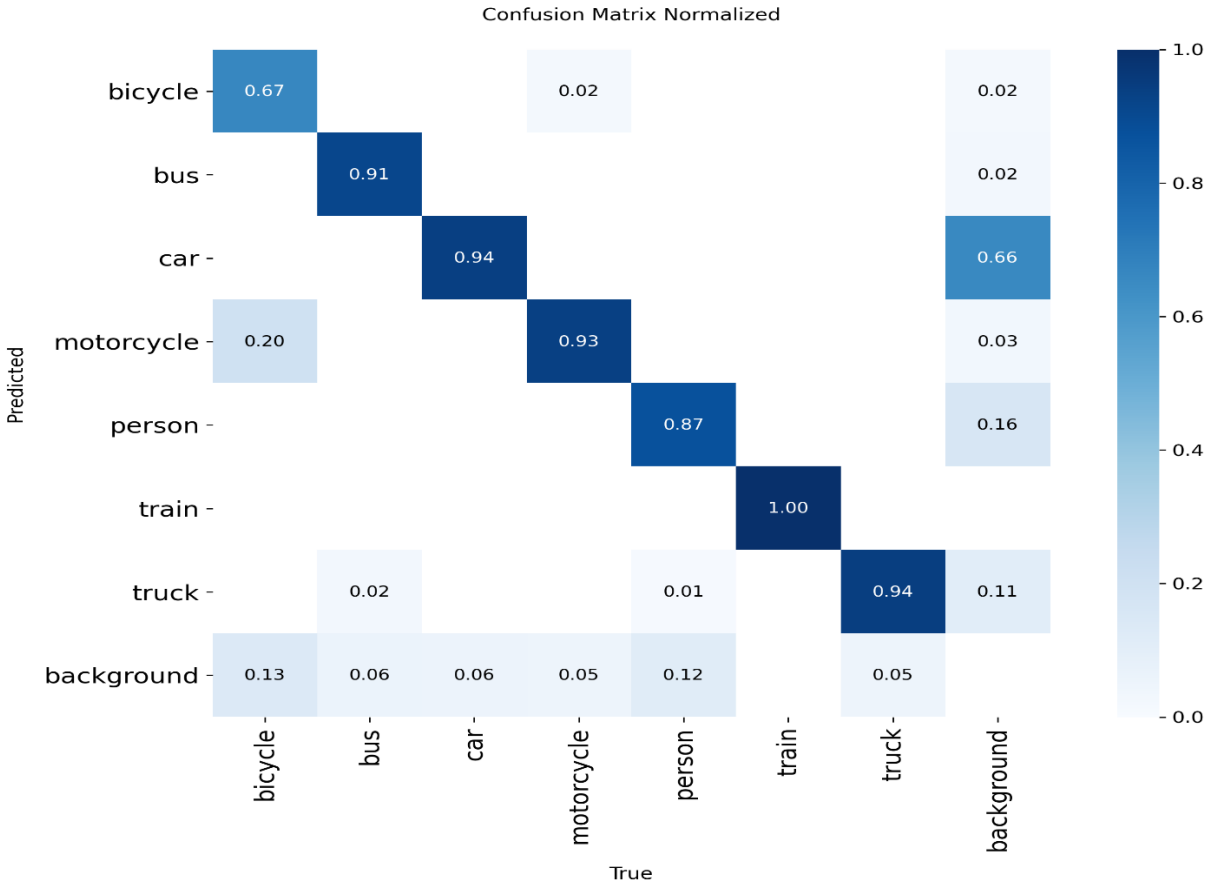


Fig. 4.30: Confusion matrix (normalized) graph for YOLOv12x

YOLOv12x demonstrates high recall but moderate precision. While it detects many objects, it also has a higher false positive rate compared to top models.

4.2.7 Performance Metrics and Graphs for YOLOv8x

Below is the summary of model evaluation based on precision, recall, F1-score, accuracy, and mAP metrics. The performance graph for the YOLOv8x model is included to visualize the training and validation performance during the experiments.

Table 4.7: YOLOv8x performance analysis

Model	Precision	Recall	F1-Score	Accuracy	mAP@50	mAP@50-95
YOLOv8x	93.64%	91.96%	92.80%	86.56%	96.14%	80.44%

Graphs for YOLOv8x:

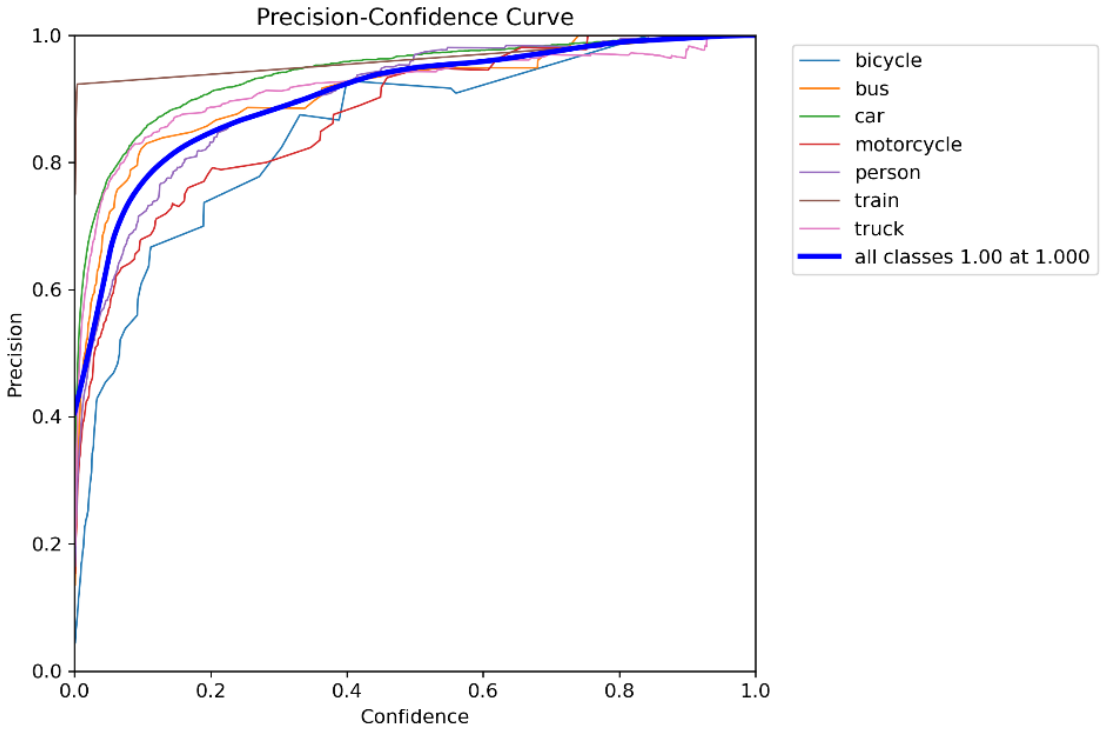


Fig. 4.31: Precision graph for YOLOv8x

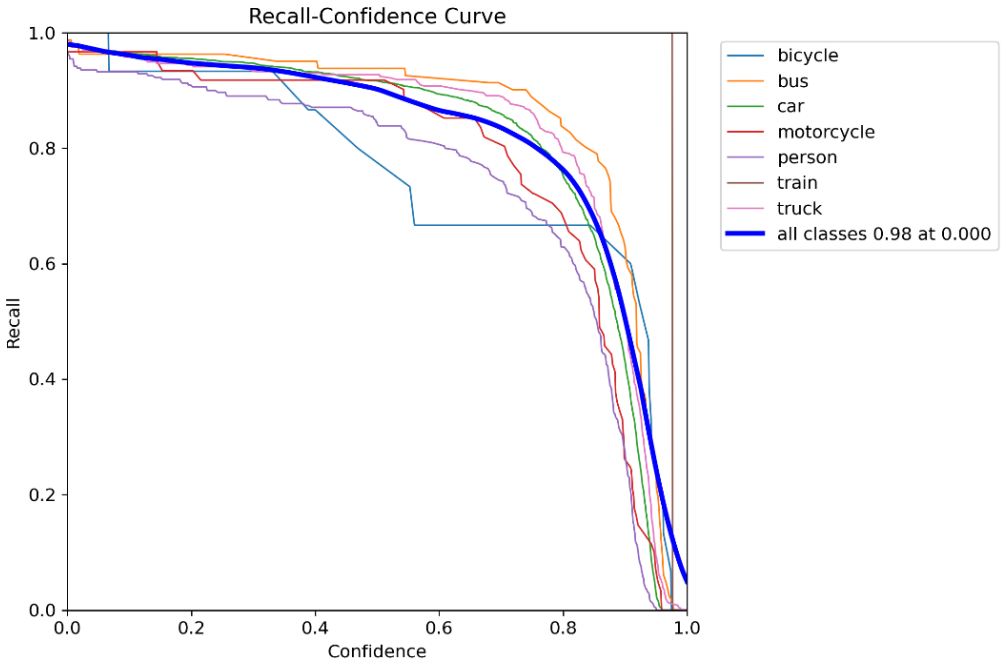


Fig. 4.32: Recall graph for YOLOv8x

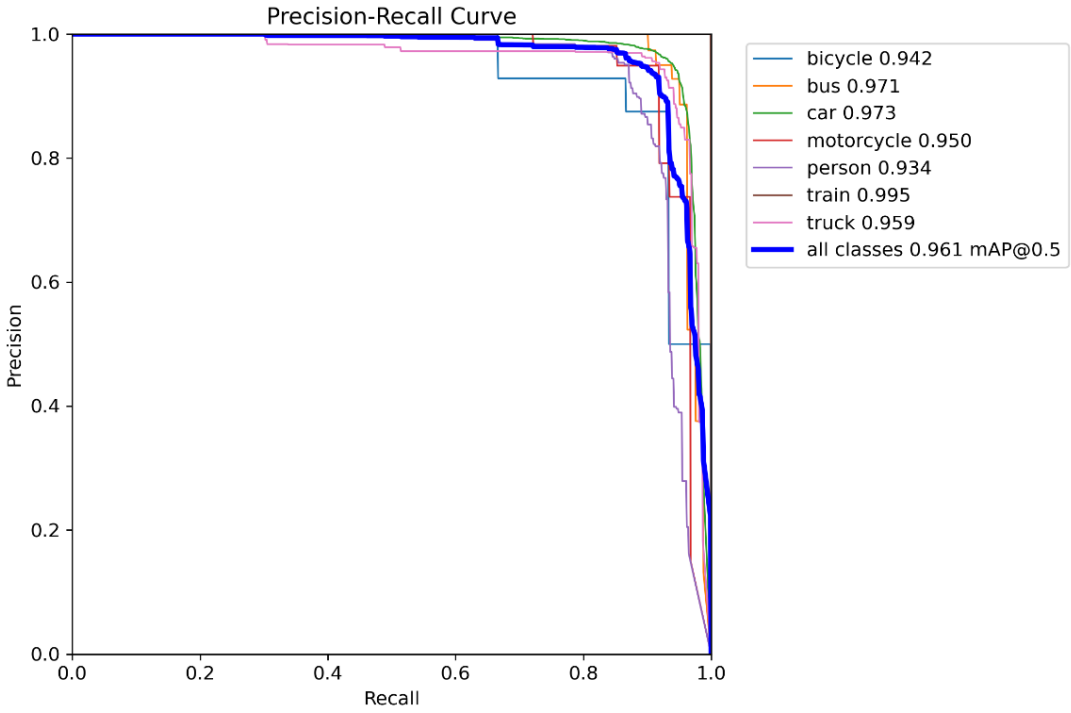


Fig.4.33: Precision-Recall graph for YOLOv8x

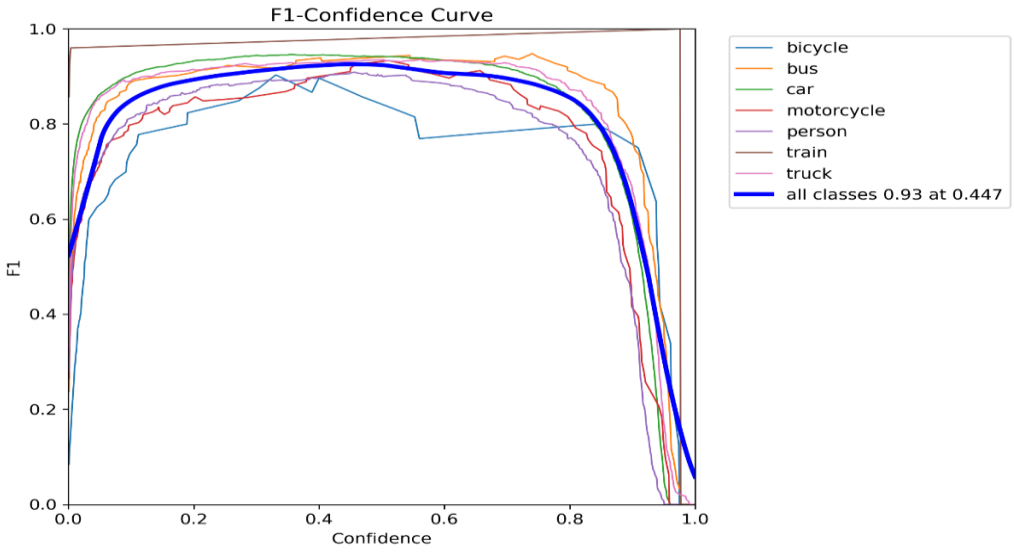


Fig.4.34: F1-score graph for YOLOv8x

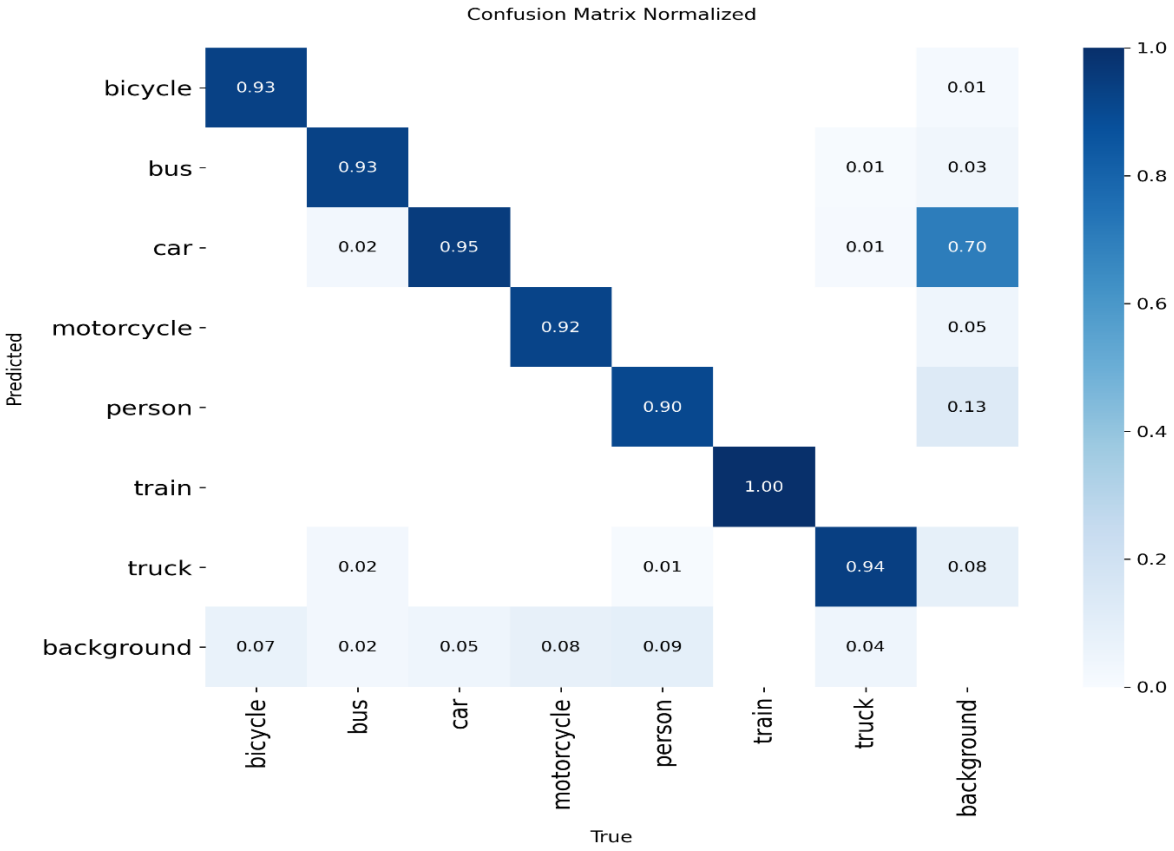


Fig.4.35: Confusion matrix (normalized) graph for YOLOv8x

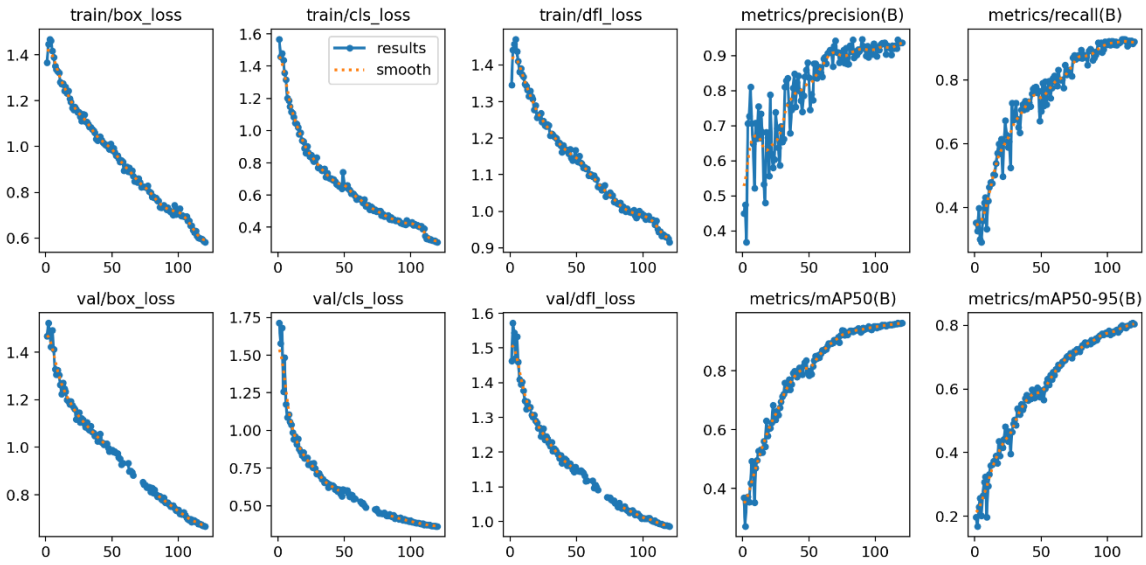


Fig. 4.36: Result graph for YOLOv8x

YOLOv8x demonstrates the best overall performance, with the highest precision, recall, F1-score, and mAP. The training and validation loss curves show near-perfect convergence, indicating excellent generalization and robustness to adverse weather conditions. The model excels across all evaluation metrics.

4.3 Comparison

In this section, we compare the performance of different YOLO models based on the previous evaluation results. The goal is to identify each model's strengths and weaknesses, understand the trade-offs, and assess their suitability for real-world applications. We will first discuss the model performance in detail, followed by a state-of-the-art comparison.

4.3.1 Model Performance

The following table summarizes the performance metrics for each YOLO model tested in this study. The models were evaluated across precision, recall, F1-score, accuracy, mAP@50, and mAP@50-95. Based on these metrics, we will analyze each model's relative strengths and weaknesses.

Table 4.8: All YOLO models performance analysis

Model	Precision	Recall	F1-Score	Accuracy	mAP@50	mAP@50-95
YOLOv11n	87.21%	79.93%	83.41%	67.95%	85.92%	65.27%
YOLOv11s	**94.57%**	71.61%	81.51%	68.79%	94.84%	72.94%
YOLOv11m	93.10%	76.03%	83.70%	71.97%	95.87%	74.20%
YOLOv11l	93.17%	90.61%	91.87%	84.96%	95.40%	77.16%
YOLOv11x	93.36%	90.91%	92.12%	85.39%	96.08%	76.63%
YOLOv12x	91.70%	87.34%	89.47%	80.94%	94.30%	74.58%
YOLOv8x	93.64%	91.96%	92.80%	86.56%	96.14%	80.44%

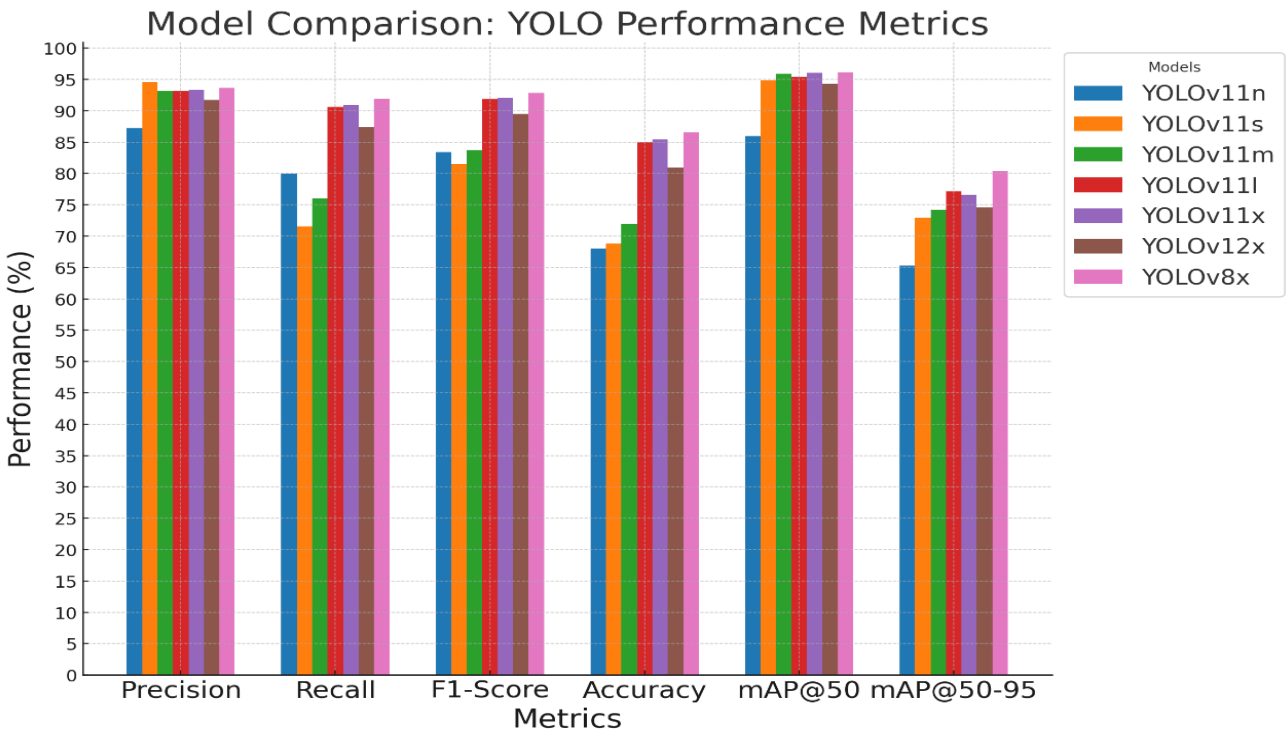


Fig. 4.37: YOLO models result comparison

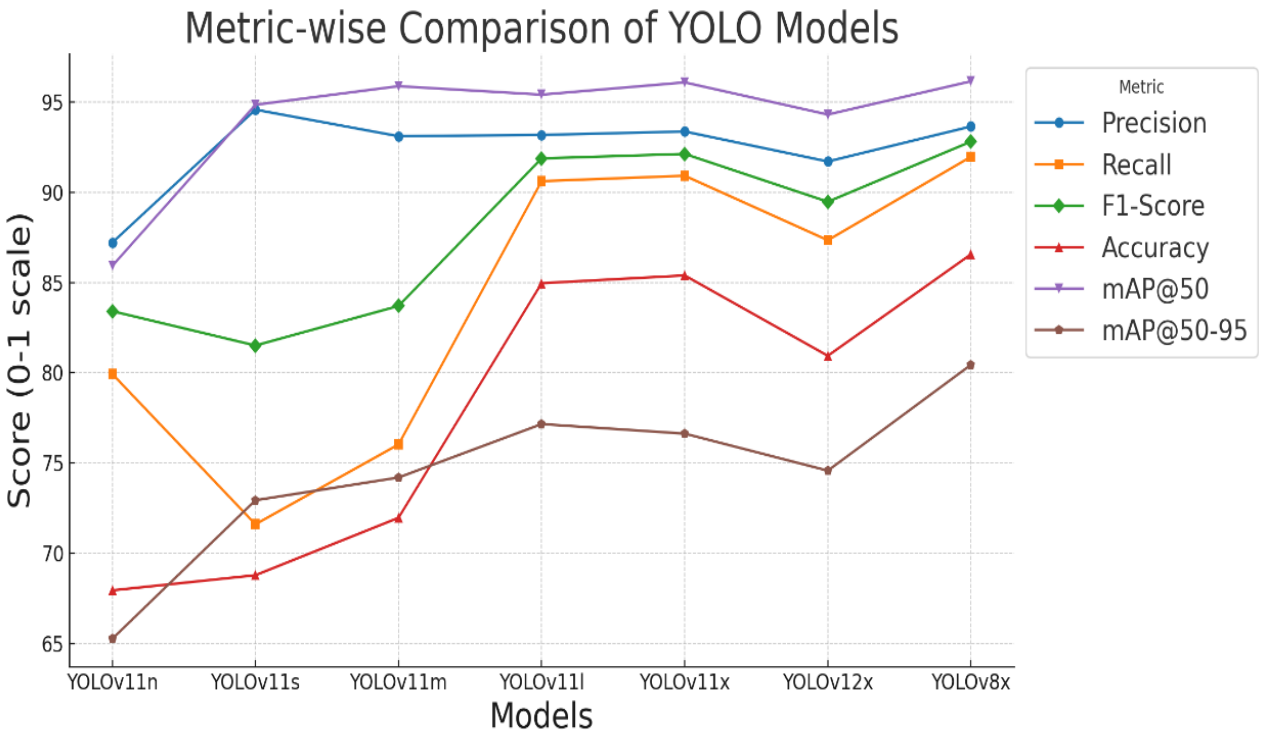


Fig. 4.38: YOLO models metric-wise result comparison

4.3.2 State of the Art

The YOLO-based models show strong performance in real-world applications, particularly with real-time inference and high accuracy under adverse weather conditions. While exact performance values for Faster R-CNN, EfficientDet, and RetinaNet on the DAWN dataset aren't available, a generalized comparison can be made based on similar tasks and datasets.

i) **Precision and Recall Comparison:** YOLOv8x outperforms many state-of-the-art models in precision (93.64%) and recall (91.96%). Faster R-CNN often has lower precision due to complex post-processing steps, resulting in more false positives, but typically has higher recall. YOLOv8x offers a balanced performance, making it ideal for environments like autonomous driving.

ii) **mAP (Mean Average Precision):** YOLOv8x achieves a high mAP@50 of 96.14%, demonstrating accurate localization and classification. Models like Faster R-CNN and

EfficientDet generally show lower mAP scores, especially in the mAP@50-95 metric, with YOLOv8x excelling in this area (80.44%).

iii) **Inference Speed:** YOLO models, especially YOLOv8x, are faster and more efficient, with inference times of 35ms per image, making them suitable for real-time applications. In contrast, models like Faster R-CNN and EfficientDet have slower inference speeds due to their more complex architectures.

iv) **Generalization to Adverse Weather Conditions:** YOLOv8x YOLOv8x stands out for its exceptional robustness when deployed in adverse weather conditions like fog, rain, and snow, making it a prime choice for autonomous vehicles operating in unpredictable environments. Unlike models such as Faster R-CNN and EfficientDet, which often face challenges with degraded image quality and low visibility in such scenarios, YOLOv8x maintains high precision and recall. This enables it to effectively detect and classify objects even when visibility is severely compromised, a critical advantage in real-world autonomous driving, where environmental factors can hinder the performance of other detection systems.

In this study, YOLOv8x was tested on the DAWN dataset, which was augmented to simulate various weather conditions. The results showed that YOLOv8x consistently outperformed other models, especially in scenarios where images were obscured by rain, fog, or snow. While Faster R-CNN and EfficientDet excel in clear conditions, they tend to underperform in low-visibility environments, often producing blurred or incomplete object detections and higher rates of false negatives.

YOLOv8x's architecture, optimized for both speed and precision, enables it to quickly and accurately detect objects in challenging weather. Its high precision reduces false positives, while its high recall ensures reliable detection, even when visibility is limited. This balance between precision and recall is crucial for the safety of autonomous vehicles, as accurate and timely object detection can prevent collisions and improve hazard detection.

Chapter 5

Social and Environmental Influence

This chapter examines the social and environmental impacts of deploying autonomous vehicles with advanced object detection systems like the YOLO models evaluated in this study. We analyze how autonomous driving can influence society, the environment, and sustainability, focusing on societal impact, environmental footprint, and the sustainability of these technologies.

5.1 Societal Impact

The advent of autonomous vehicles (AVs) and technologies like YOLO object detection systems is set to reshape modern society. As AVs move from experimental to fully operational, they will significantly impact the economy, public health, legal frameworks, and social norms. This section discusses the societal implications, exploring both positive and negative aspects of these changes.

5.1.1 Financial and Health Influences

The financial impact of autonomous vehicles (AVs) is significant, offering efficiency gains and cost reductions in sectors like transportation and logistics. AVs could lower operational costs by replacing human drivers, offering 24/7 service, and potentially lowering transportation prices for consumers. However, the high initial investment required for AV development could limit accessibility, and industries reliant on human labor, like trucking and taxi services, may face job losses, increasing income inequality.

From a health perspective, AVs could reduce traffic accidents caused by human error, leading to fewer injuries and fatalities. This would lower medical costs and reduce

healthcare system burdens. However, mental health concerns could arise from increased reliance on automation, potentially affecting drivers' decision-making skills. Data privacy issues also raise concerns about surveillance and privacy breaches as AVs collect vast amounts of data.

5.1.2 Safety, Legal, and Cultural Issues

While AVs promise to reduce accidents caused by human error, they introduce new risks, including technical failures or sensor malfunctions. These risks need to be addressed through rigorous testing and robust systems to ensure vehicle reliability. Legally, questions of liability arise if an AV causes an accident, requiring updates to legal frameworks and the development of new insurance models.

Cultural acceptance of AVs varies, with some societies viewing it as a loss of personal freedom, while others see it as a solution to traffic problems. Public trust in AV technology will be a key factor in its adoption. Additionally, ethical concerns regarding data privacy and hacking of AVs need to be addressed to ensure passenger safety and privacy.

5.2 Impact on the Environment

The environmental impact of autonomous vehicles (AVs) is a key consideration. While AVs offer potential environmental benefits, challenges exist in their production, energy consumption, and end-of-life disposal.

i) **Environmental Benefits:** Autonomous vehicles (AVs), especially when combined with electric vehicle (EV) technology, can reduce greenhouse gas emissions and air pollution. They improve fuel efficiency, reduce traffic congestion, and optimize driving behavior, leading to lower fuel consumption. AVs can also improve traffic flow through smart routing and communication, further reducing emissions. The adoption of electric AVs would significantly lower the transportation sector’s carbon footprint.

ii) **Environmental Concerns:** Despite the benefits, AV production introduces environmental challenges, such as the energy-intensive manufacturing of sensors, batteries, and AI systems. The extraction of materials for lithium-ion batteries, like lithium, cobalt, and nickel, causes environmental degradation. Additionally, training AI models like YOLO requires substantial computational resources, contributing to carbon emissions unless powered by renewable energy sources.

iii) **Mitigation Strategies:** To mitigate environmental impacts, manufacturers and governments should focus on improving battery technology, such as solid-state or recyclable batteries. Using renewable energy for AI model training and operating AV fleets can further reduce the carbon footprint of these technologies.

5.3 Sustainability Issues

Sustainability is a key challenge in the adoption of autonomous vehicles (AVs). While AVs can help address environmental concerns, their long-term sustainability depends on factors like manufacturing, lifecycle management, and the energy required for operation.

i) **Resource Usage and Supply Chain:** A key sustainability challenge is the demand for raw materials needed to manufacture autonomous vehicles (AVs), such as sensors and batteries. Materials like lithium and cobalt are sourced from regions with problematic mining practices, raising ethical and environmental concerns. Ensuring sustainable sourcing will be critical as AV adoption increases.

ii) **Life-Cycle Emissions and End-of-Life Disposal:** While AVs, especially electric ones, are more environmentally friendly than gasoline vehicles, they still contribute to life-cycle emissions, particularly during production. At the end of their life, proper recycling of AVs, batteries, and components is essential to reduce environmental impact. Improved recycling systems are needed to handle the growing number of AVs.

iii) **Long-Term Sustainability:** The long-term sustainability of autonomous vehicles (AVs) hinges on their integration with existing infrastructure, optimization of energy

usage, and provision of affordable transportation. As AVs continue to evolve, they must be incorporated into urban planning and transportation systems in a way that complements existing infrastructure while reducing congestion and enhancing efficiency. Collaboration among urban planners, policymakers, and engineers is essential to create supporting infrastructure, such as smart traffic systems, dedicated AV lanes, and electric vehicle (EV) charging stations. Energy optimization is vital for AV sustainability, particularly with the increasing shift towards clean energy and electric vehicles (EVs) to reduce carbon emissions. Governments must incentivize EV production and adoption through policies like tax breaks, subsidies, and investments in charging infrastructure. Managing energy consumption efficiently is crucial, especially considering the energy-intensive nature of training AI models and operating AVs in real-time. Affordable transportation is another cornerstone of AV sustainability. To realize the full potential of AVs, the technology must be accessible to all. While the initial cost of AVs may be high, technological advancements and economies of scale should eventually reduce costs, making them affordable for the average consumer. AVs also offer the potential for lower transportation costs, particularly in shared mobility applications, where costs can be distributed across multiple users. However, there are significant challenges, particularly in the environmental impact of AV production. Manufacturing electric AVs requires raw materials like lithium, cobalt, and nickel for batteries, the extraction of which can cause environmental harm, human rights issues, and ecological damage. To mitigate these effects, innovations in battery technology are necessary, alongside responsible sourcing of raw materials. Additionally, developing circular economies for AV batteries, such as recycling or reusing them, will help reduce environmental damage. Another challenge is managing waste at the end of the vehicle's life.

Chapter 6

Complex Engineering Problems and Activities

This chapter explores the engineering challenges involved in designing, implementing, and deploying autonomous vehicles (AVs) equipped with YOLO-based object detection models. It highlights technical difficulties, integration issues, and system complexities that need to be addressed for effective AV deployment. The chapter focuses on challenges such as sensor integration, real-time object detection, data processing, and model optimization to ensure safe and efficient AV operation. Additionally, it examines the engineering activities, including system design, model training, data augmentation, and algorithm optimization, used to overcome these challenges.

6.1 Addressing Complex Engineering Problems

Autonomous vehicles (AVs) represent one of the most advanced applications of AI and ML, offering improvements in road safety, traffic efficiency, and transportation systems. However, the complexity of AV systems presents engineering challenges across sensor integration, real-time processing, AI model training, and system optimization. The integration of object detection technologies like YOLO models adds further complexity, as these models must function effectively in dynamic, real-world environments.

1. Sensor Integration and Multi-Sensor Fusion

A critical engineering challenge in autonomous vehicles (AVs) is integrating multiple sensor systems, including cameras, LiDAR, radar, and ultrasonic sensors. Each sensor provides different data, making it necessary to fuse these data streams into an accurate representation of the vehicle's environment. Advanced algorithms, like Kalman filters and Bayesian networks, are used for sensor fusion to handle noise and uncertainties, especially in low-visibility conditions, ensuring safe navigation.

2. Real-Time Processing and Inference Speed

AVs must process large amounts of sensor data quickly for real-time decision-making. YOLO-based models are known for fast inference but balancing model complexity with real-time performance is challenging. Techniques like model compression, quantization, pruning, and edge computing help reduce inference times and improve real-time processing efficiency, essential for time-sensitive applications like autonomous driving.

3. AI Model Training and Generalization

Training YOLO models for AVs requires diverse and well-labeled datasets, which are time-consuming to create. Data augmentation techniques simulate different weather and lighting conditions to expose the model to varied environments. Transfer learning and careful hyperparameter tuning ensure the model generalizes well and can handle unforeseen real-world scenarios.

4. Handling Adverse Weather and Low-Visibility Conditions

Adverse weather, such as fog, rain, snow, and sandstorms, severely affects sensor performance, especially cameras and LiDAR. Weather-specific data augmentation and multi-modal sensor fusion (e.g., combining radar, LiDAR, and cameras) help improve object detection under poor visibility conditions. Training with weather-specific data ensures AVs perform well even in challenging weather.

5. System Integration and Communication

AV systems must integrate sensors, AI models, and control systems into a cohesive unit. Communication protocols ensure smooth data flow from sensors to AI models, and from models to the vehicle's control system. Real-time processing, minimal latency, and fault tolerance are critical for safety. Redundancy in system design ensures AVs can safely handle component failures.

6.2 Addressing Complex Engineering Activities

The development and deployment of autonomous vehicles (AVs) with object detection technologies like YOLO require complex engineering activities throughout the vehicle's lifecycle, from initial design to ongoing maintenance. These activities involve technical tasks that ensure the vehicle operates efficiently, safely, and robustly in unpredictable real-world conditions.

This section explores key engineering activities required for successful development, testing, and deployment of autonomous driving systems, covering system design, data processing, AI model training, real-time implementation, and post-deployment monitoring. Proper execution of these activities ensures AV reliability and performance in real-world applications.

1. System Design and Architecture

System design is a critical engineering activity in autonomous vehicle (AV) development, involving the integration of sensors, AI models, control systems, and communication protocols. The design must ensure seamless operation for real-time decision-making in complex environments. Key factors in the design include:

i) **Sensor Integration:** Engineers integrate sensors like cameras, LiDAR, and radar, using fusion techniques to process data and create an accurate representation of the surroundings.

ii) **Control Systems and Decision-Making:** The control system makes driving decisions based on model outputs (e.g., bounding boxes and object classes), translating predictions into actions like steering, accelerating, or braking.

iii) **Communication Protocols**: Engineers design protocols for fast, secure data flow between sensors, AI models, and control systems, minimizing latency for timely decision-making.

Once the architecture is defined, engineers optimize and integrate individual components to ensure they function cohesively.

2. Data Acquisition and Augmentation

Data acquisition is crucial for training effective object detection models. Engineers gather large datasets representing diverse driving environments, including road types, traffic conditions, and weather scenarios. The DAWN dataset serves as an example, containing labeled images of cars, trucks, and pedestrians. However, collecting real-world data is time-consuming, so data augmentation is used to artificially expand the dataset. Techniques like adding fog, rain, snow, and altering lighting conditions help models generalize better to real-world environments. Synthetic data generation through simulation can also create labeled training data for extreme conditions, making it useful when acquiring real-world data is impractical.

3. Model Training and Optimization

Once data is acquired and augmented, training the object detection models begins. This process involves selecting the appropriate model architecture (e.g., YOLO variants), designing the loss function, tuning hyperparameters, and performing cross-validation to optimize performance.

i) **Model Selection:** YOLO variants (e.g., YOLOv8x, YOLOv11n, YOLOv11s) are chosen for their high-speed, high-accuracy detection. Trade-offs between inference speed and computational cost are considered based on the system's requirements.

ii) **Loss Function Design:** A multi-task loss function is used to optimize both classification and localization tasks, ensuring effective convergence and good generalization to unseen data.

iii) **Hyperparameter Tuning:** Key hyperparameters like learning rate, batch size, and epochs are optimized using techniques such as grid search or random search to achieve the best performance.

iv) **Cross-validation:** Cross-validation techniques like k-fold or hold-out validation are used to prevent overfitting and ensure the model generalizes well to new data.

v) **Optimization Algorithms:** Optimization algorithms such as Adam, AdamW, or SGD are employed to update model weights and minimize the loss function during training.

4. Real-Time Model Deployment and Testing

After training, the object detection model is deployed in a real-time autonomous vehicle system, integrated with the vehicle's control systems to interpret predictions and make real-time decisions. Real-time inference requires quick processing while maintaining accuracy. Techniques like quantization, pruning, and edge computing are used to optimize the model, reducing computational load and speeding up inference. This ensures quick decision-making for tasks such as object detection, tracking, and avoidance.

i) **Sensor Integration:** The model is integrated with the vehicle's sensor systems (e.g., cameras, LiDAR, radar), allowing real-time data reception and accurate predictions to guide vehicle actions.

ii) **Testing and Validation:** Extensive testing is conducted in both controlled environments (e.g., test tracks) and real-world scenarios to ensure the model can handle diverse conditions, including low light and adverse weather like rain, snow, or fog.

5. Post-Deployment Monitoring and Maintenance

After deployment, autonomous vehicle performance must be continually monitored to ensure optimal operation. Post-deployment activities include:

i) **Performance Evaluation:** Engineers regularly assess the model's performance by collecting data from the vehicle's sensors and comparing detection accuracy with expected results. If performance degrades over time or in specific conditions, adjustments or retraining may be necessary.

ii) **Updating the Model:** As new real-world data is gathered, the model can be retrained or fine-tuned to improve performance, ensuring the system adapts to evolving driving conditions.

iii) **Safety and Compliance:** In addition to monitoring performance, safety and compliance are critical aspects of post-deployment monitoring. Autonomous vehicles must adhere to local traffic laws and safety standards, and these regulations can vary across regions and evolve over time. Ensuring that the AV's software and systems remain compliant with the latest regulations is paramount for legal operation and public safety. Routine safety checks must be conducted on all vehicle systems, including the object detection models, the vehicle's navigation and control systems, and the data collection infrastructure. Software updates are performed regularly to patch any security vulnerabilities and protect the system from emerging cyber threats. These updates may include improvements in detection algorithms, bug fixes, or security enhancements to guard against potential hacking attempts or malicious software.

Moreover, regulatory bodies may introduce new safety requirements for autonomous vehicles as the technology advances. It is crucial to stay ahead of these changes by ensuring that the vehicle's systems comply with the latest safety standards. For instance, if new regulations require stricter standards for pedestrian detection, the model may need to be updated or retrained to meet these enhanced criteria. Ongoing compliance ensures that AVs remain safe to operate, protect passengers and pedestrians, and maintain the integrity of the vehicle's safety features.

Chapter 7

Conclusion and Future Plan

This chapter summarizes the work carried out in this thesis, highlighting the results achieved and contributions to the development of autonomous vehicle technologies using YOLO-based object detection models. It also discusses the study's limitations and outlines future plans for enhancing the system's capabilities.

7.1 Conclusion

This thesis explored the use of YOLO-based object detection models to enhance autonomous vehicles (AVs) in navigating complex environments, especially under adverse weather conditions. The goal was to evaluate how various YOLO models perform in real-world scenarios like fog, rain, snow, and sandstorms, which can impair perception systems. The evaluation showed that while all YOLO variants performed well under clear conditions, smaller models like YOLOv11n and YOLOv11s struggled in adverse weather. Larger models such as YOLOv8x, YOLOv11x, and YOLOv12x showed superior robustness to environmental distortions like occlusion, motion blur, and lighting changes.

Key findings from the study include:

i) **Model Selection and Adaptation:** Larger models provided higher accuracy but required more computational power, highlighting the trade-off between speed and accuracy.

ii) **Adverse Weather Training:** Data augmentation with simulated weather conditions improved model robustness, with models trained on diverse, high-quality datasets performing better.

iii) **Performance Optimization:** Techniques like quantization, pruning, and edge computing were crucial for maintaining inference speed while ensuring detection accuracy.

The results of this work significantly contribute to autonomous driving by offering a framework for evaluating real-time object detection under various adverse conditions. The findings provide insights into optimizing AI-based systems, particularly object detection models, for safety-critical applications in real-world driving environments. While the models demonstrated potential in autonomous vehicle applications, further optimization is necessary to ensure high performance in diverse, complex conditions. The insights gained offer a roadmap for improving object detection systems in the evolving field of autonomous driving.

7.2 Limitations

This study provides valuable insights into the performance of YOLO-based object detection models for autonomous vehicles, but several limitations were encountered that highlight areas for future improvement.

1. Dataset Limitations

The DAWN dataset, while diverse, was limited in size and lacked edge cases like rare object types or extreme weather conditions. A more comprehensive dataset would improve model performance and generalization.

2. Limited Adverse Weather Simulation

The simulated weather conditions (fog, rain, snow, and sandstorms) may not fully capture the complexities of real-world environments, as these conditions can vary dynamically in different regions. Static augmentation methods may not replicate real-world variability.

3. Computational Constraints

Training and inference times were considerable, especially for larger YOLO variants, due to limited computational resources on Kaggle's free GPUs. Access to more powerful hardware would enable more comprehensive training and evaluations.

4. Lack of Real-World Testing

The study relied mostly on simulated environments. Real-world testing is crucial to address practical challenges such as sensor noise, vehicle dynamics, and changing road conditions, which were not fully represented.

5. Model Generalization and Overfitting

Despite cross-validation efforts to reduce overfitting, the models may still have become too specialized to the DAWN dataset. Additionally, more advanced architectures, such as transformer-based models or ensemble learning, could have improved performance in extreme conditions.

7.3 Future Plan

Building on the insights and results from this study, several promising areas for future research and development can further improve the performance of object detection systems for autonomous vehicles, particularly in challenging environments:

i) Expanding the Dataset

A key direction is to enhance the DAWN dataset or develop a new one with more diverse environmental conditions, including rare events and edge cases. This would involve gathering data across different weather, lighting, and traffic scenarios. Real-time data from actual autonomous vehicles, combined with sensor fusion from cameras, LiDAR, radar, and other sensors, would further enrich the dataset and improve model robustness.

ii) Improved Adverse Weather Simulation

Future work should aim for dynamic and realistic simulations of adverse weather by incorporating real-time data from external sources like weather stations or satellite imagery. Enhancing synthetic data generation to simulate extreme weather, such as rapid fog movement or heavy rain, will improve object detection under these challenging conditions.

iii) Integration with Advanced Sensor Fusion Systems

A multi-sensor approach will be crucial for future AV systems. Integrating data from various sensors (LiDAR, radar, ultrasonic) will enable better object detection, especially in low-visibility environments. Developing robust real-time sensor fusion frameworks that combine these inputs will be essential for enhancing the safety and reliability of AVs.

iv) Advanced Model Architectures and Transfer Learning

Exploring advanced architectures like transformer-based models or attention mechanisms can improve handling complex scenes and object occlusions. Additionally, transfer learning can be employed to fine-tune models on smaller, environment-specific datasets, reducing the need for large-scale training data and improving model efficiency.

v) Real-World Testing and Evaluation

Real-world testing should be prioritized to assess model performance in actual driving scenarios, accounting for factors such as vehicle speed, road conditions, and sensor calibration. Real-time feedback from field vehicles will enable engineers to refine the models, ensuring their reliability and effectiveness in real-world applications.

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Appendix

A. Dataset Information

i) DAWN Dataset

The DAWN dataset is utilized for training and evaluating YOLO-based object detection models in adverse weather conditions, such as fog, rain, snow, and sandstorms.

Classes: 7 classes (Bicycle, Bus, Car, Motorcycle, Person, Train, Truck).

Data Split:

Training Set: 974 images

Validation Set: 425 images

Test Set: Initially empty; populated using a 70/20/10 split rule from training and validation data.

ii) Weather Augmentation

Data augmentation techniques are used to simulate weather conditions and increase model robustness:

a. Fog Simulation: Reduces visibility and introduces blur.
b. Rain Simulation: Adds water droplets to the lens, causing occlusion.
c. Snow Simulation: Introduces snowflakes, causing motion blur.
d. Sandstorm Simulation: Adds dust particles, resulting in significant noise and blur.

B. Model Architectures

i) YOLOv8x Architecture

a. Backbone: CSPDarknet for efficient feature extraction.
b. Neck: PANet (Path Aggregation Network) for multi-scale feature fusion.
c. Head: Multi-task loss for bounding box, classification, and objectness prediction.

ii) YOLOv11 (n, s, m, l, x) Variants

a. Backbone: Variations of CSPDarknet, optimized for different computational requirements.

b. Neck: PANet and FPN (Feature Pyramid Network) for better feature aggregation.
c. Head: Multi-task loss function for real-time predictions.

iii) YOLOv12x

a. Backbone: Enhanced architecture with multi-modal sensor fusion for better performance under extreme weather conditions.
b. Neck: Optimized PANet and FPN for better handling of occlusions.
c. Head: Incorporates self-attention mechanisms for more precise object detection.

C. Hyperparameter Settings

Optimizer: AdamW with learning rate schedules.

Training Details:

Epochs: 120 epochs for each model variant.

Batch Size: 4.

Learning Rate: 0.000909.

Loss Function: Combination of objectness loss, classification loss, and bounding box loss.

D. Model Evaluation

i) Evaluation Metrics:

a. Precision: Measures how many of the predicted objects were correct.
b. Recall: Measures how many of the actual objects were detected.
c. F1-Score: A balanced metric combining precision and recall.
d. Accuracy: Overall correctness of the model.
e. mAP (mean Average Precision): Average precision at various intersection-over-union (IoU) thresholds.

ii) Performance Results:

a. The models were evaluated on their ability to detect objects under adverse weather conditions, and their performance was compared across various weather simulations (fog, rain, snow, sandstorms).

b. YOLOv8x showed the best performance in terms of accuracy, while smaller models like YOLOv11n performed well in less complex environments with faster inference times.

E. Additional Results and Figures

i) Performance Graphs: Graphical representations of precision, recall, F1-score, accuracy, and mAP values for each model variant across different weather conditions.

ii) Confusion Matrices: Visualizations of model prediction accuracy for each class (Bicycle, Bus, Car, etc.).

F. Model Training and Tuning Results

i) Optimization Algorithms: Hyperparameter optimization was performed using Gray Wolf Optimization (GWO) and Clever Optimization (CLEO) algorithms. These optimizers adjusted the learning rate, batch size, layers, and activation functions to maximize performance while maintaining real-time processing capabilities.

ii) Training Configuration: All models were trained for 120 epochs on the augmented and enhanced dataset. Training was done using GPUs (NVIDIA T4) on a Kaggle notebook environment.

G. Code and Implementation Details

i) Code for Dataset Preprocessing: Code used for data augmentation and preprocessing of the DAWN dataset.

ii) Model Training Scripts: Scripts for training the YOLOv8x, YOLOv11, and YOLOv12x models, including commands for fine-tuning the models on the augmented dataset using the AdamW optimizer.